



National Engineering School of Tunis
University of Tunis El Manar

Chapter One: Introduction to Generative AI

Mariam ZAOUALI, PhD., Eng.
mariam.zaouali@enit.utm.tn

Powered by:



Plan



Séance 1 Chapter 1 : Introduction to Generative AI

Séance 2 Chapter 2 : Large Language Models – Hands-On

Séance 3-4 Chapter 3 : Building Applications + Certif

Séance 3-4 Hands-On : Prompt Engineering -First Assessment – Challenge revealed

Séance 5 Hands-On : LLM Ochestration – Agentic AI + Certif

Séance 6 Chapter 4 : Advanced Topics and Closing thoughts + Certif

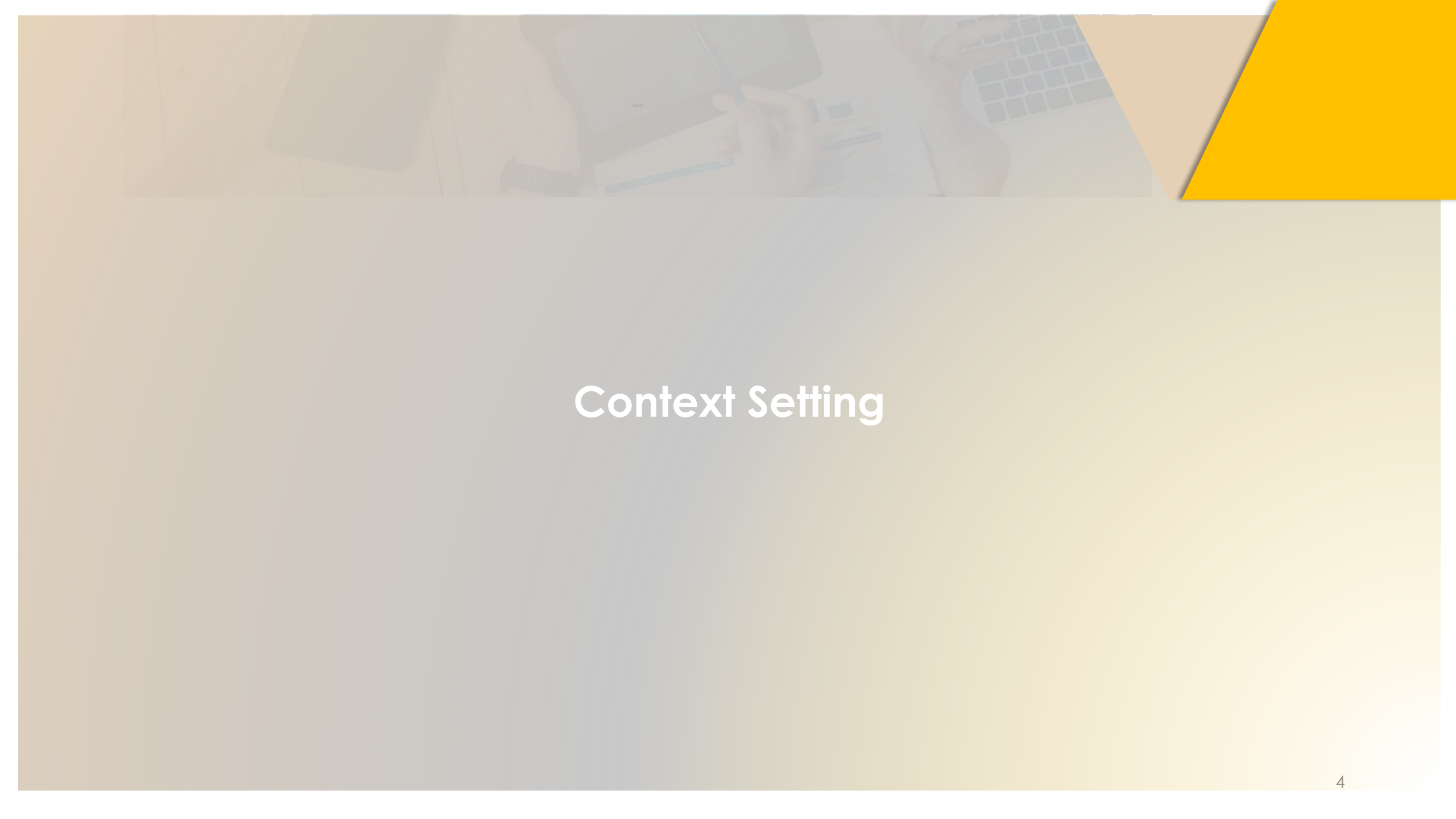
Séance 7 Challenge validation and Final Assessment

Plan



NVIDIA certifications

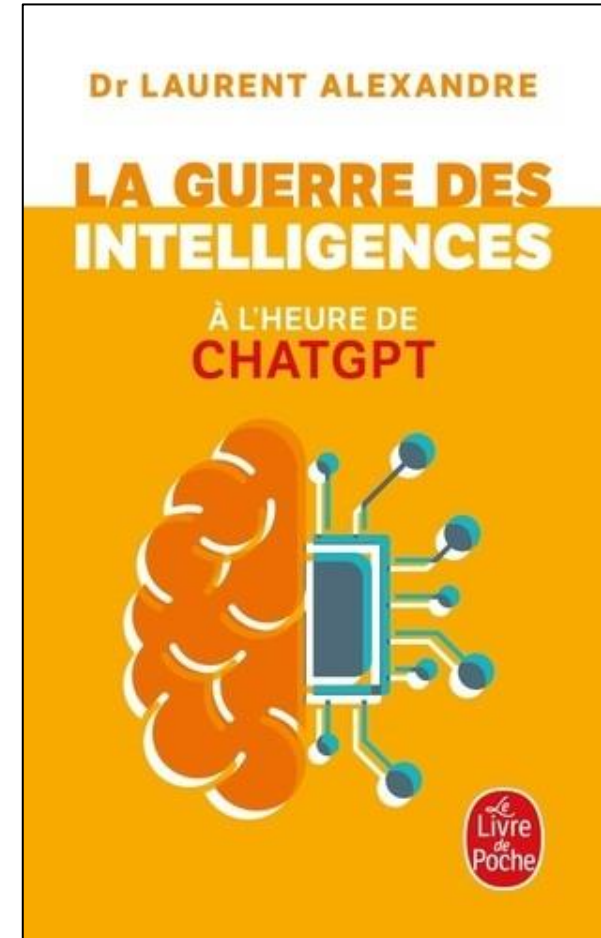
- Rapid Application Development with Large Language Models (LLMs) : [syllabus](#)
- Building Agentic AI Applications with LLMs : [syllabus](#)
- Building RAG Agents with LLMs : [syllabus](#)
- Introduction to Deploying RAG Pipelines for Production at Scale : [syllabus](#)



Context Setting

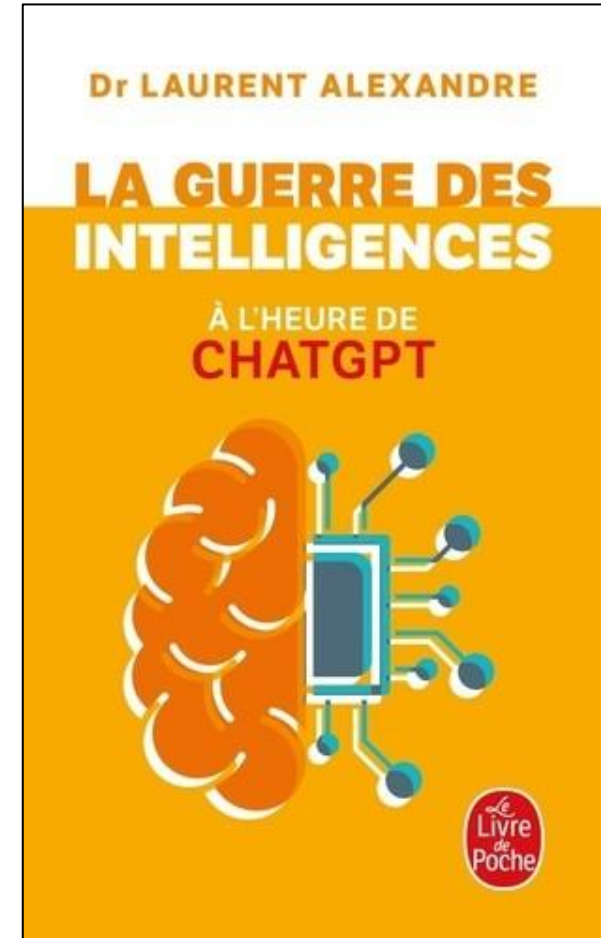
Context Setting

- **Book:** *The War of Intelligences* by Dr. Laurent ALEXANDRE (October 4th, 2017)
- **Artificial intelligence will surpass human intelligence in most cognitive domains.**
 - We must understand, regulate, and coexist with this intelligence, not ignore it.
- **The book strongly emphasizes schooling and the training of younger generations**
 - Schools must teach how to think, not just how to know.

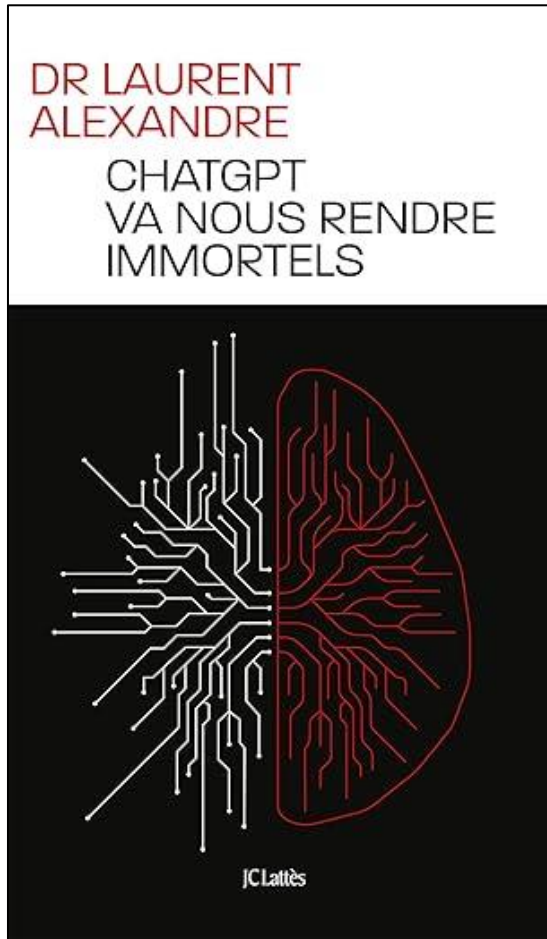


Context Setting

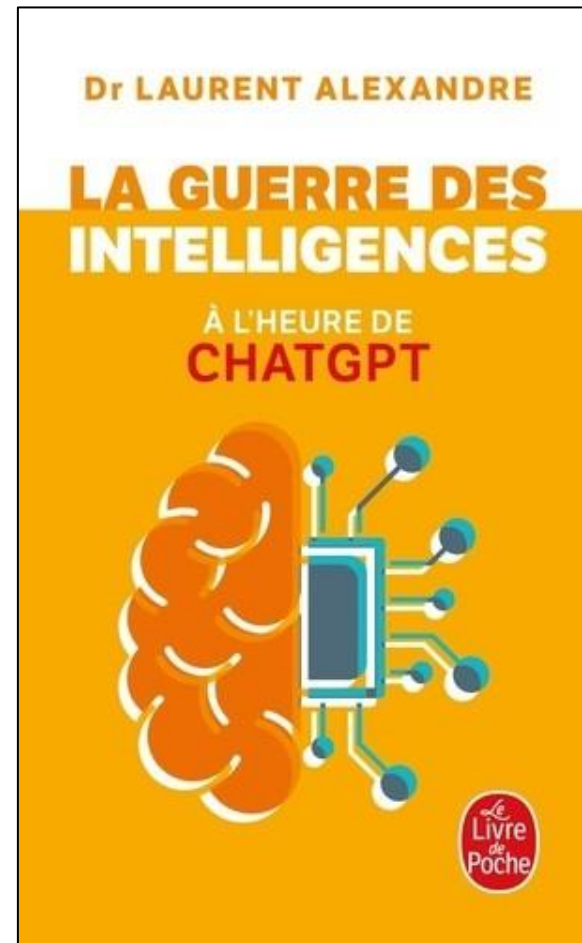
- Europe risks becoming a mere **consumer of technology**, dependent on other powers.
- Emotional intelligence and human wisdom remain **irreplaceable**.
- **Our humanity** is our best weapon in this war of intelligences



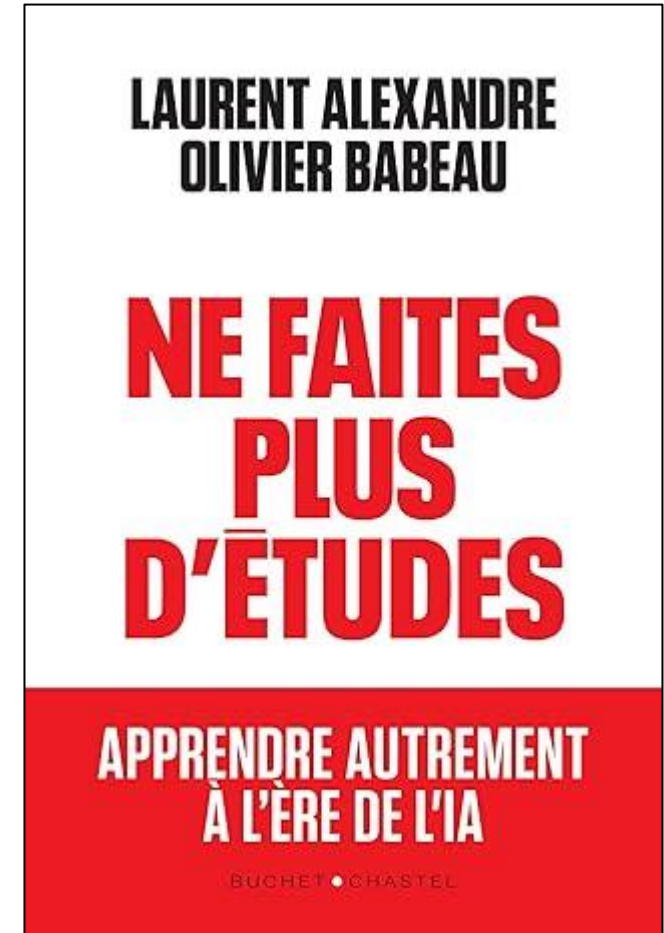
Context Setting



2011

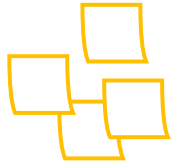


2017



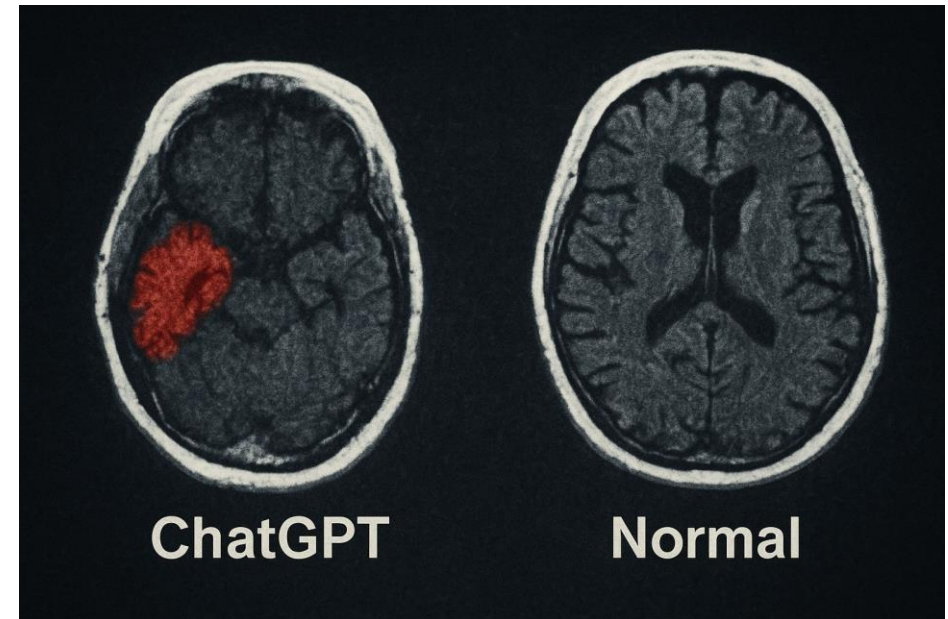
2025

If you like your brain cells, stop using ChatGPT **now!**



You don't **learn** by **feeding the prompt**. You learn by facing the **resistance**. Then, when the ideas are yours, you can use the tool **to refine, expand, and test**.

You can't outsource thinking. **Not if you want to keep it.**



John R. Saunders  • 2e

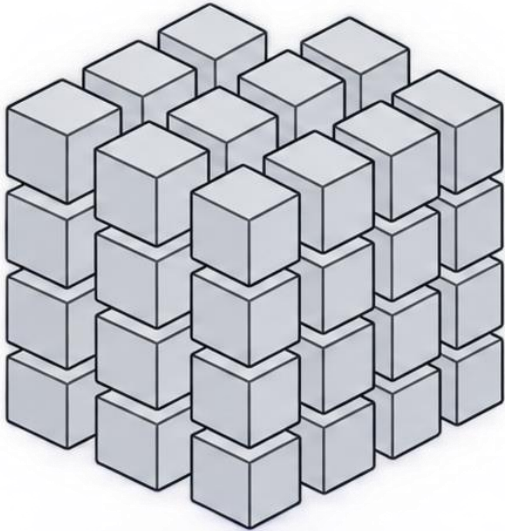
Co-Founder REBELHEAD Media | Applied Neuroscience Educator...

2 mois • Modifié • 

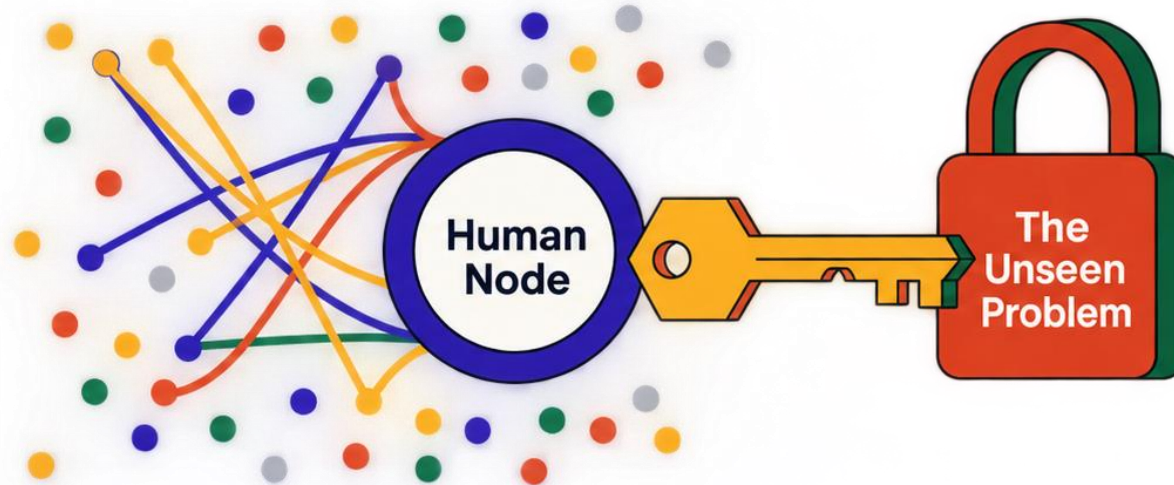
Context Setting

The Core Survival Skill: Synthesis

The Cramming Industry: Memorizing solutions to known problems.



The Synthesis Engine: Rapidly taking completely unseen information to form a novel, functional idea.

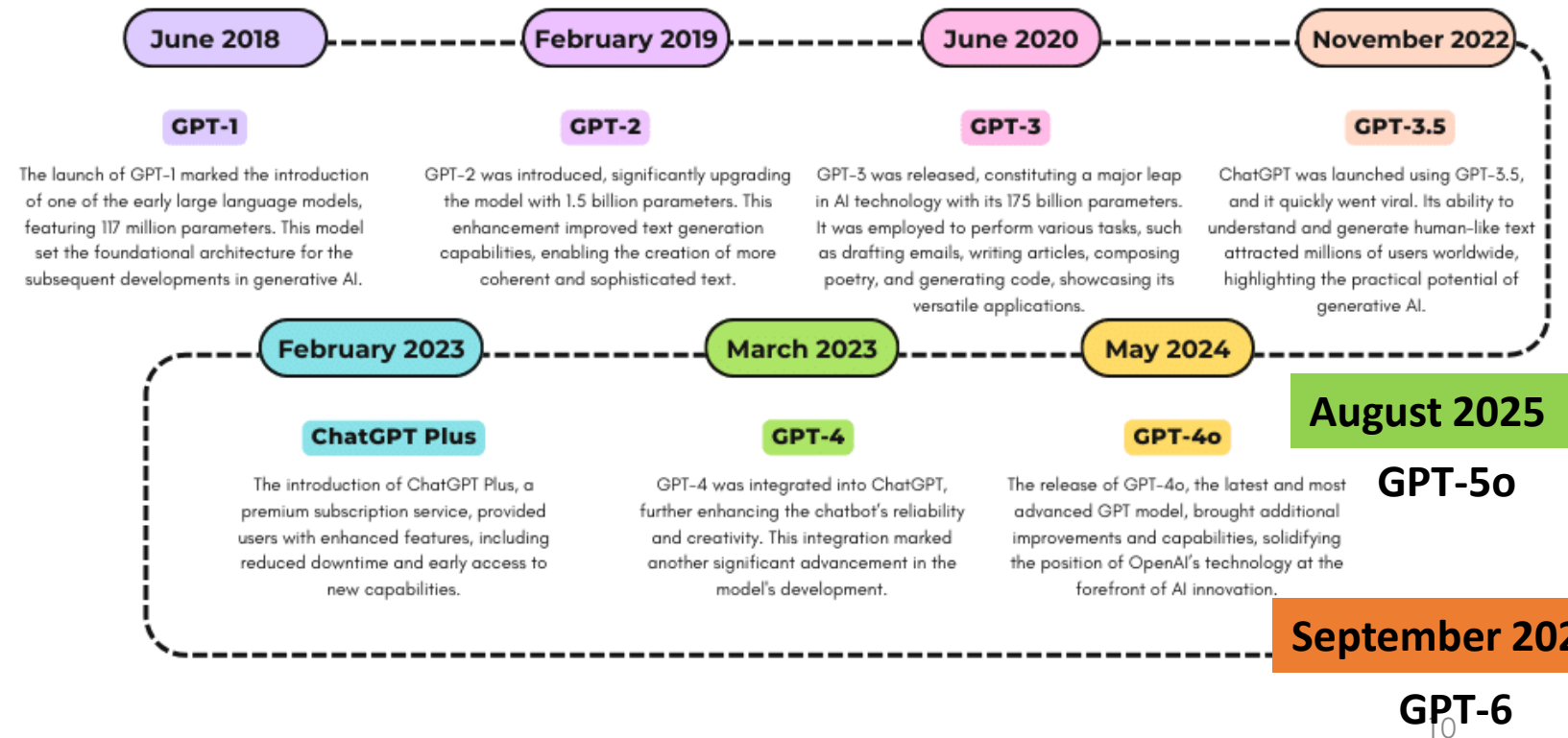


The world changes too fast for fixed curriculums. Survival demands mastering the unknown, not memorizing the known.

Context setting

ChatGPT Changed the world of knowledge forever

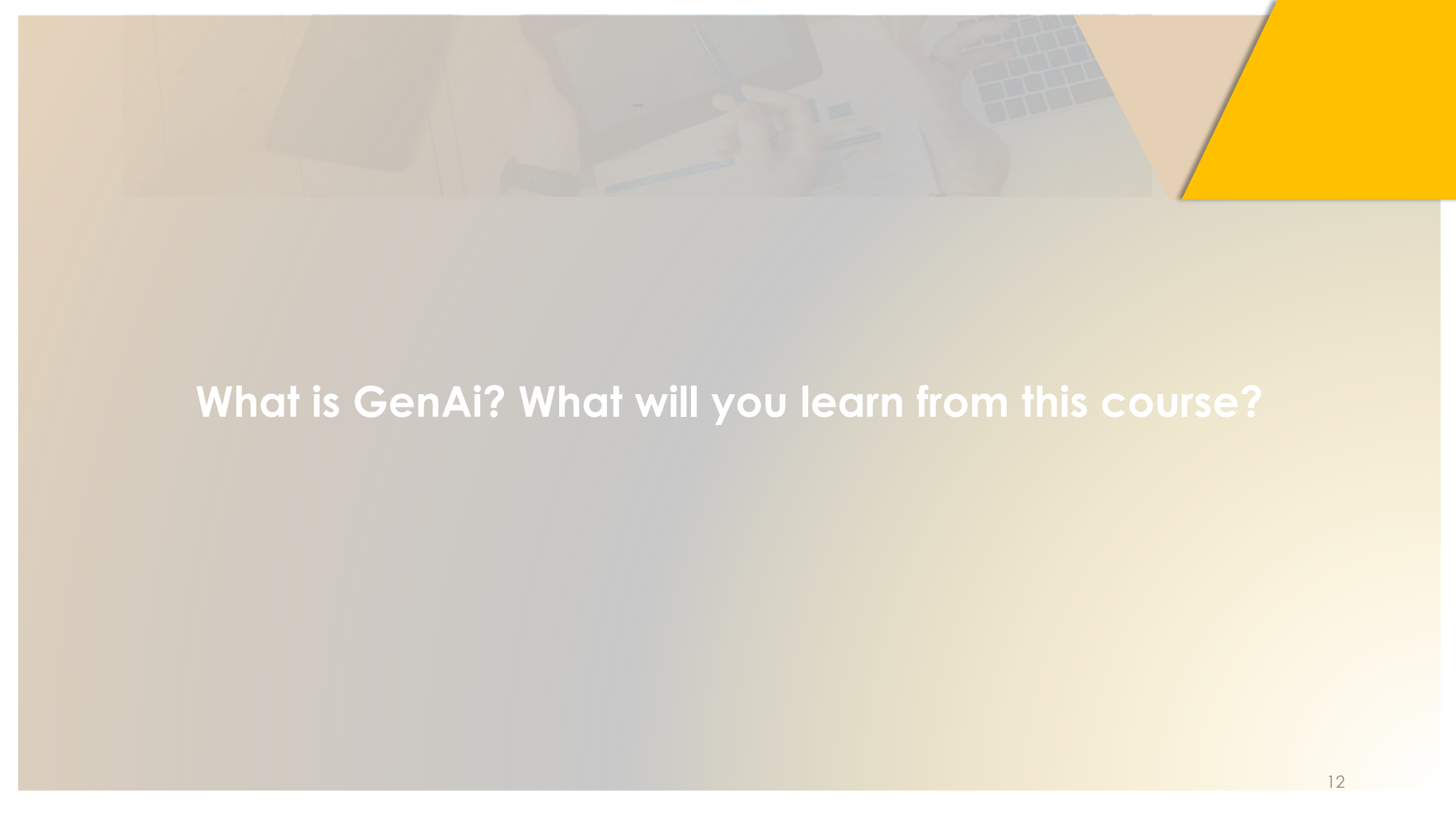
- Released Nov 2022 Took the world by storm
- How does it know so much?
- Is it “thinking” ?
- What is it made of?



Context setting

- OpenAI was a non-profit organization founded in 2015 with the goal of developing artificial **intelligence that would benefit humanity as a whole.**
- The initiative was **initially supported by Elon Musk and Sam Altman**, among others.
- When OpenAI later created a **for-profit subsidiary (ClosedAI) in 2019** and closed its code source, Elon Musk disagreed with this shift in mission **and left the board.**





What is GenAi? What will you learn from this course?

What is Gen Ai

GenAI (aka Generative AI) refers to **a subset of models** and approaches that have been trained specifically to **generate** new data.

This is different to other, un/supervised machine learning models that are trained to classify data samples, or connect input to output values for regression

Artificial Intelligence

Is the field of study

Machine Learning

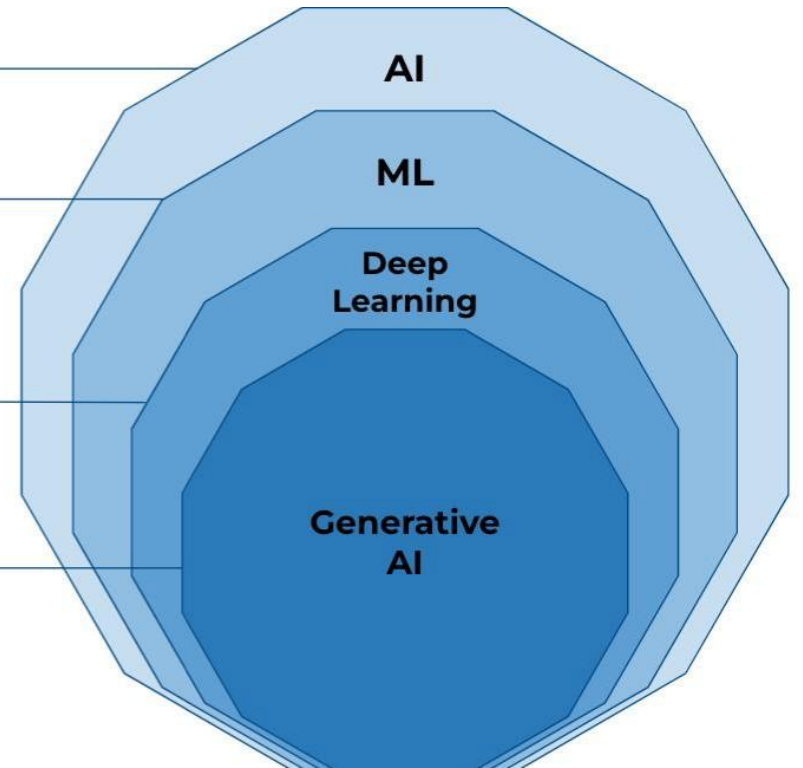
Is a branch of AI that focus on the creation of intelligent machines that learn from data. Another very well know branch inside AI is **Optimization**.

Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.
Examples: CNNs, RNNs

Generative AI

A type of ANNs that generate data that is similar to the data it was trained on.
Examples: GANs, LLMs



Word Embeddings, Tokens, and NLP

Tiktokenizer

System

You are a helpful assistant

X

User

Content

X

Add message

```
<|im_start|>system
You are a helpful assistant<|im_end|>
<|im_start|>user
<|im_end|>
<|im_start|>assistant
```

Token count
18

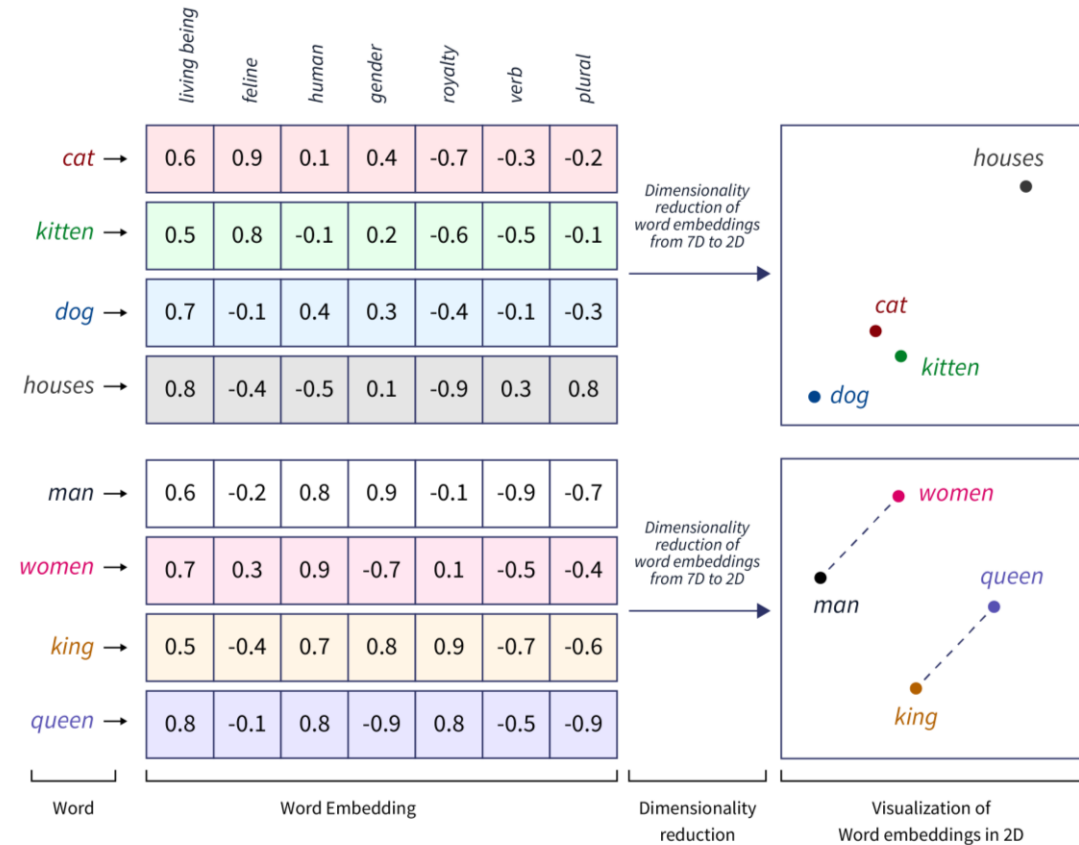
Price per prompt
\$0.000018

```
<|im_start|>system
You are a helpful assistant<|im_end|>
<|im_start|>user
<|im_end|>
<|im_start|>assistant
```

```
[100264, 9125, 198, 2675, 527, 264, 11190, 18328, 1002
65, 198, 100264, 882, 198, 100265, 198, 100264, 78191,
198]
```

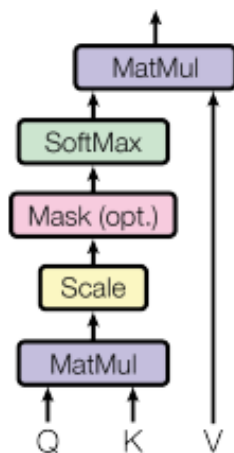
☐ Show whitespace

Built by dqbd. Created with the generous help from Diagram.

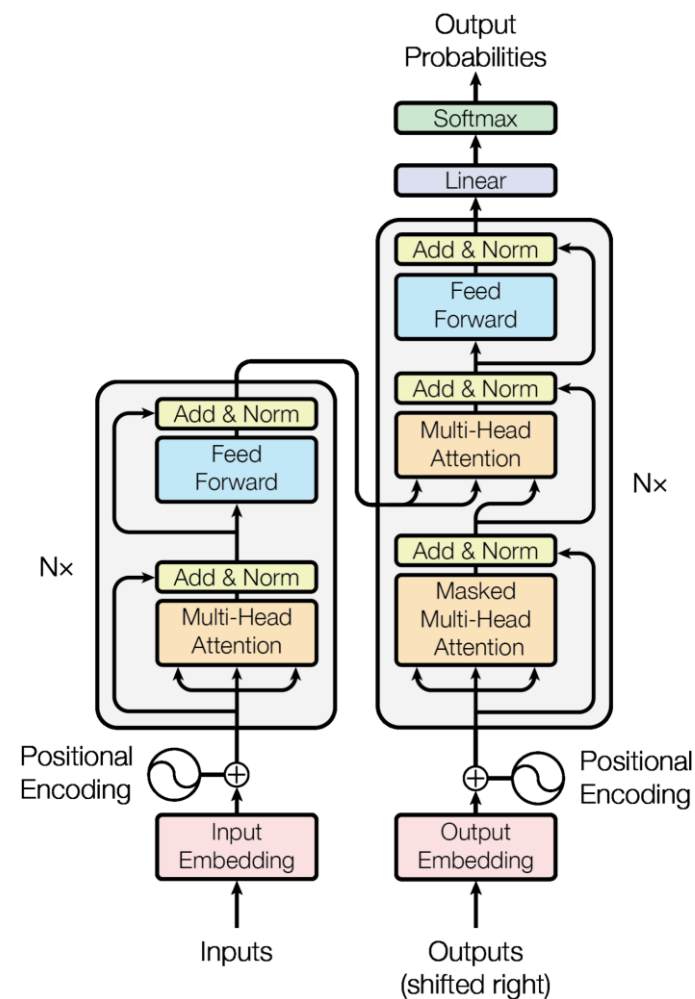
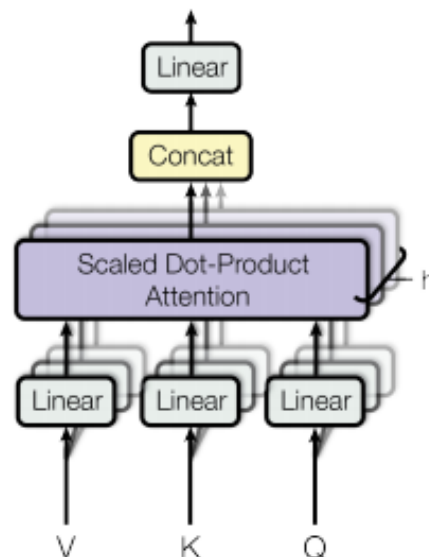


Large Language Models and the Transformer

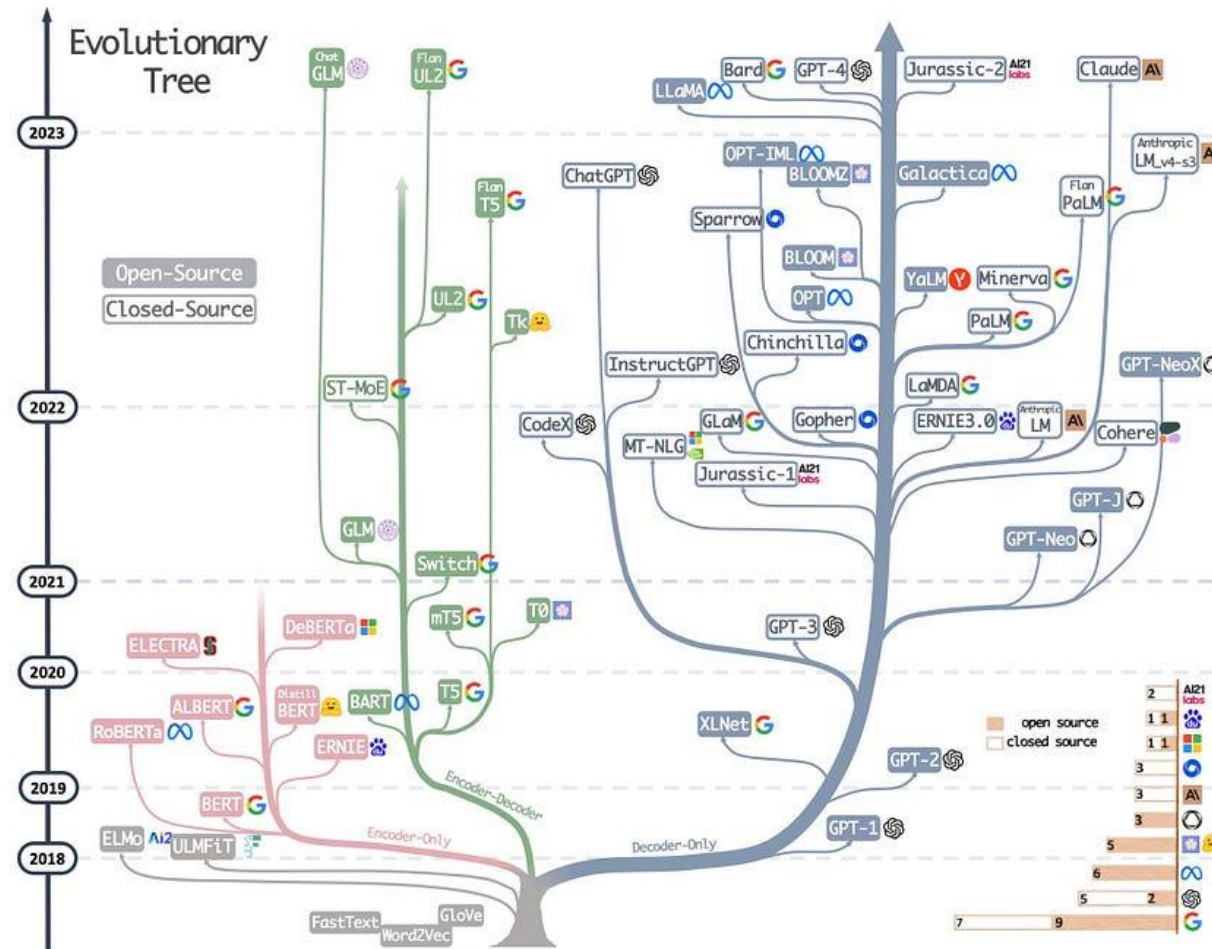
Scaled Dot-Product Attention



Multi-Head Attention

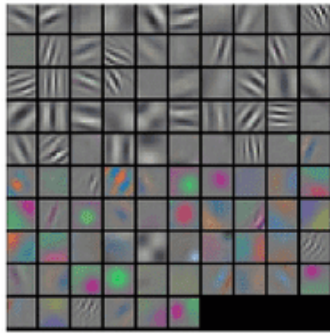


LLM Scaling Laws and LLM Families

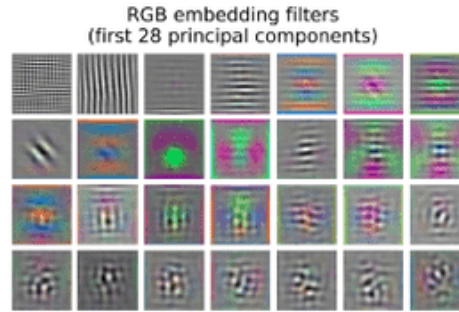


Multimodal Learning and its Applications

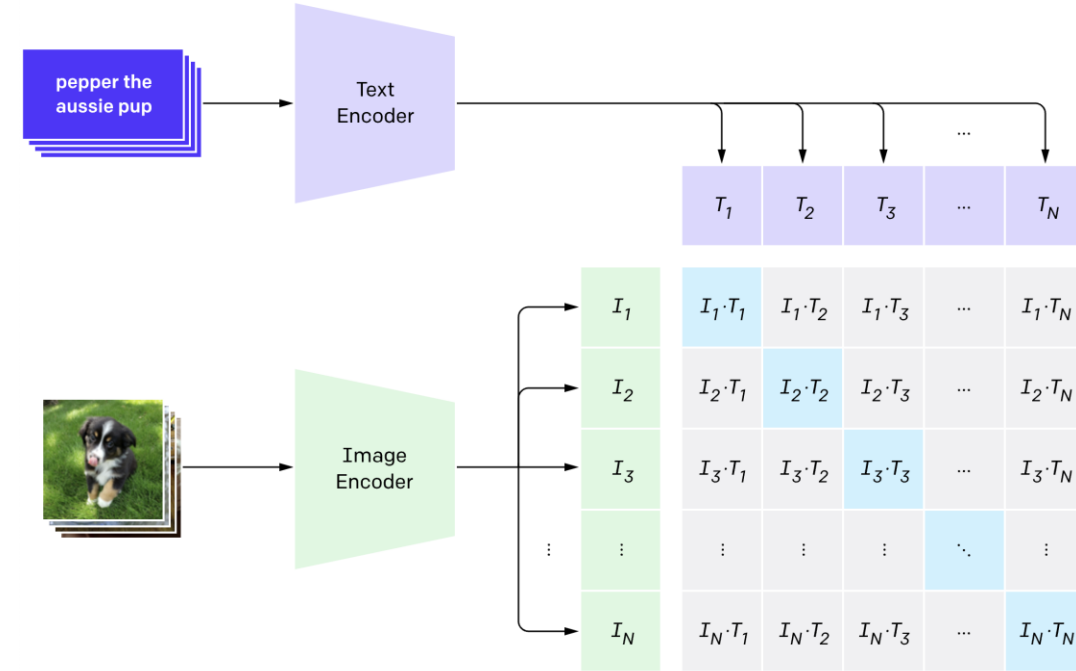
Alexnet 1st conv filters



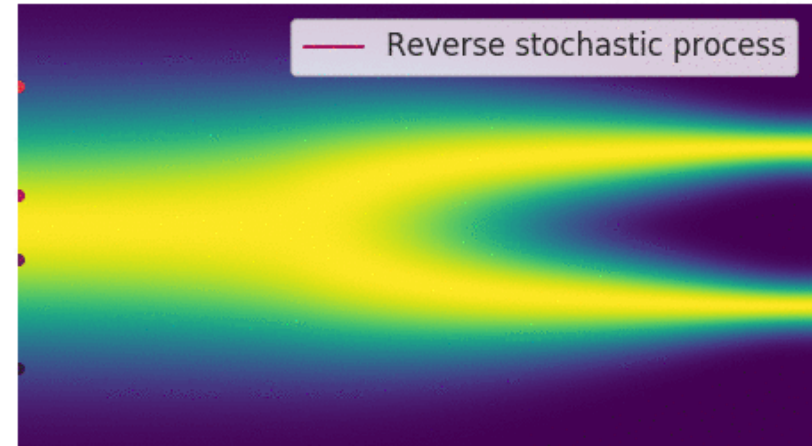
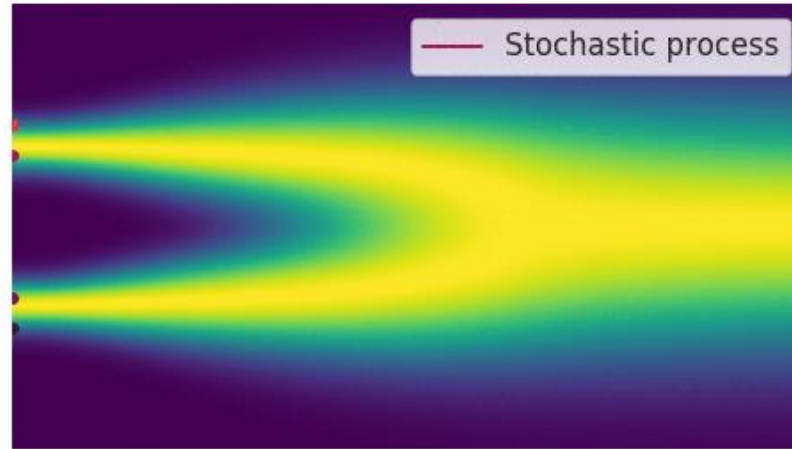
ViT 1st linear embedding filters



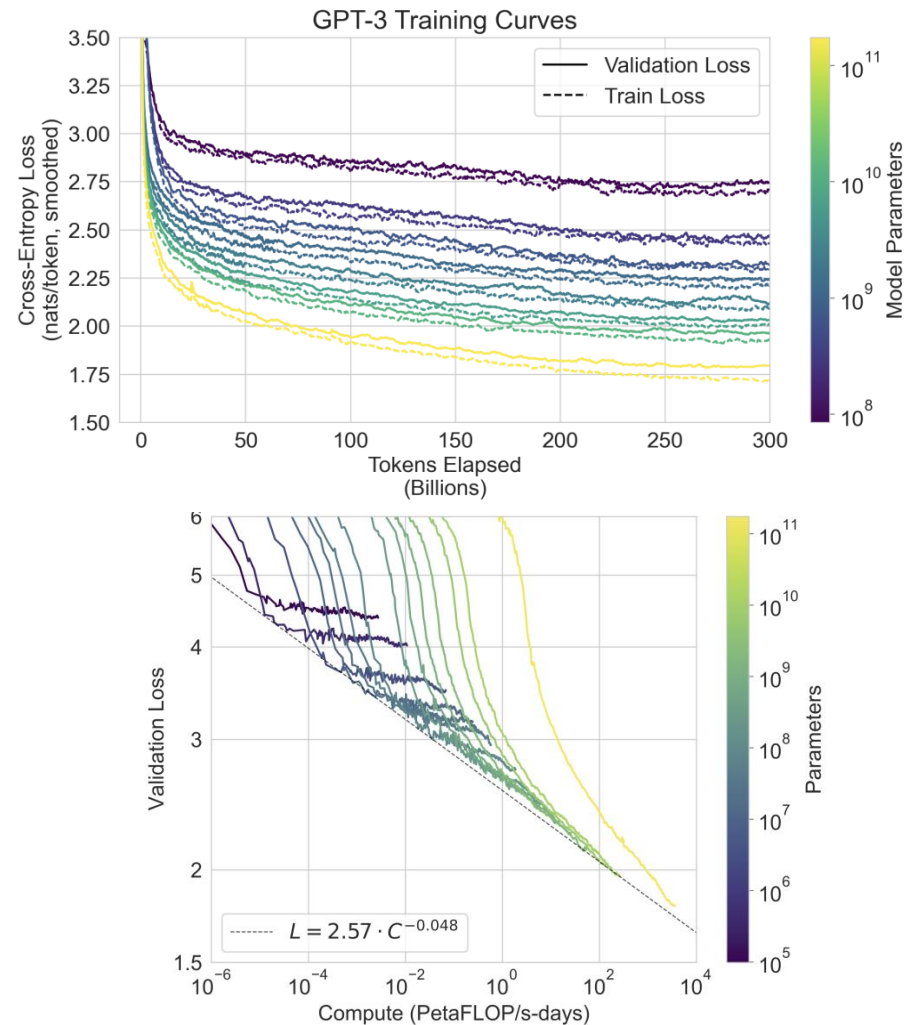
1. Contrastive pre-training



Diffusion Models in Generative AI



Model Training (Pre-Training, Instruction Follow and PEFT)



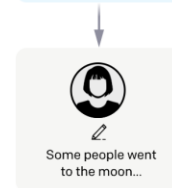
Step 1

Collect demonstration data, and train a supervised policy.

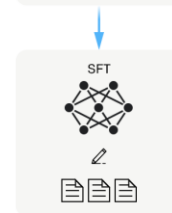
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



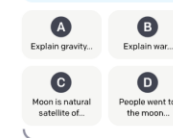
This data is used to fine-tune GPT-3 with supervised learning.



Step 2

Collect comparison data, and train a reward model.

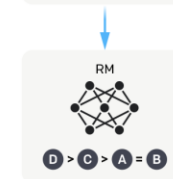
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.



The policy generates an output.

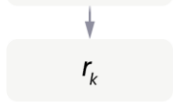


Once upon a time...

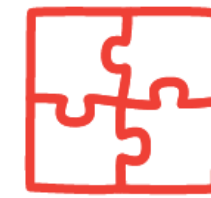
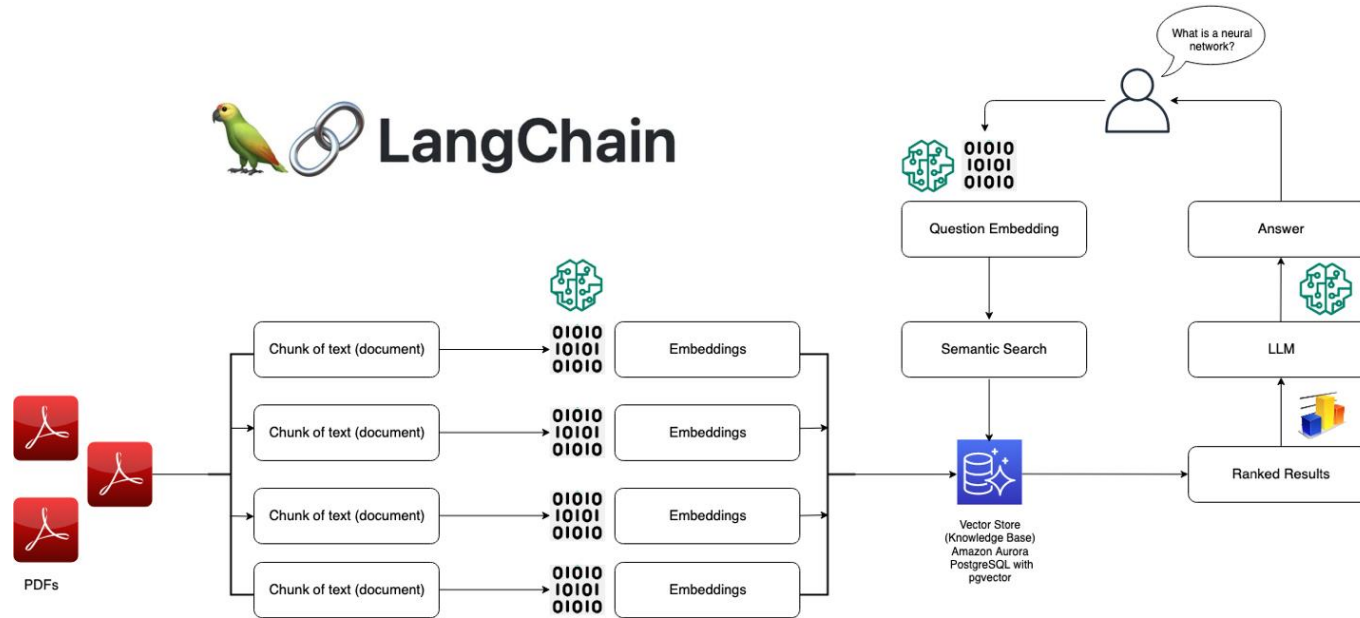
The reward model calculates a reward for the output.



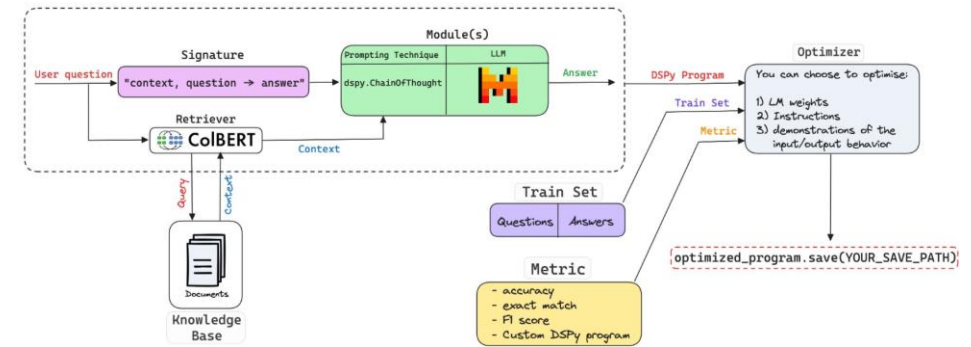
The reward is used to update the policy using PPO.

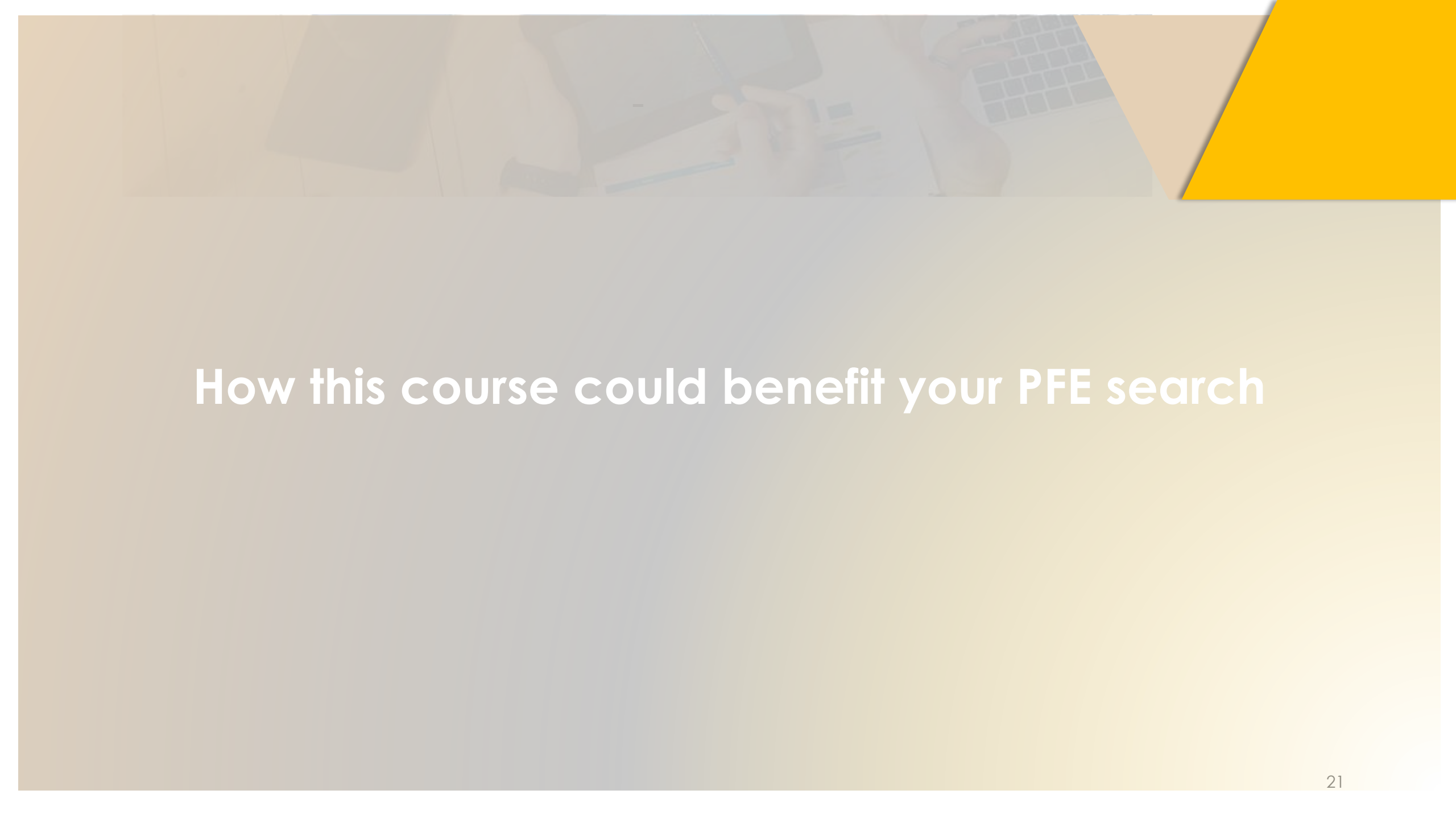


LLM Orchestration



DSPy





How this course could benefit your PFE search

PFE Trends

Agentic IA - Industriel

Développement d'un outil générique IA pour la migration de monolithes vers des micro-frontends

ArchForge : Vers une industrialisation de la conception d'architectures logicielles

AutoTARA : système multi-agents pour la gestion des risques cybernétiques des automobiles

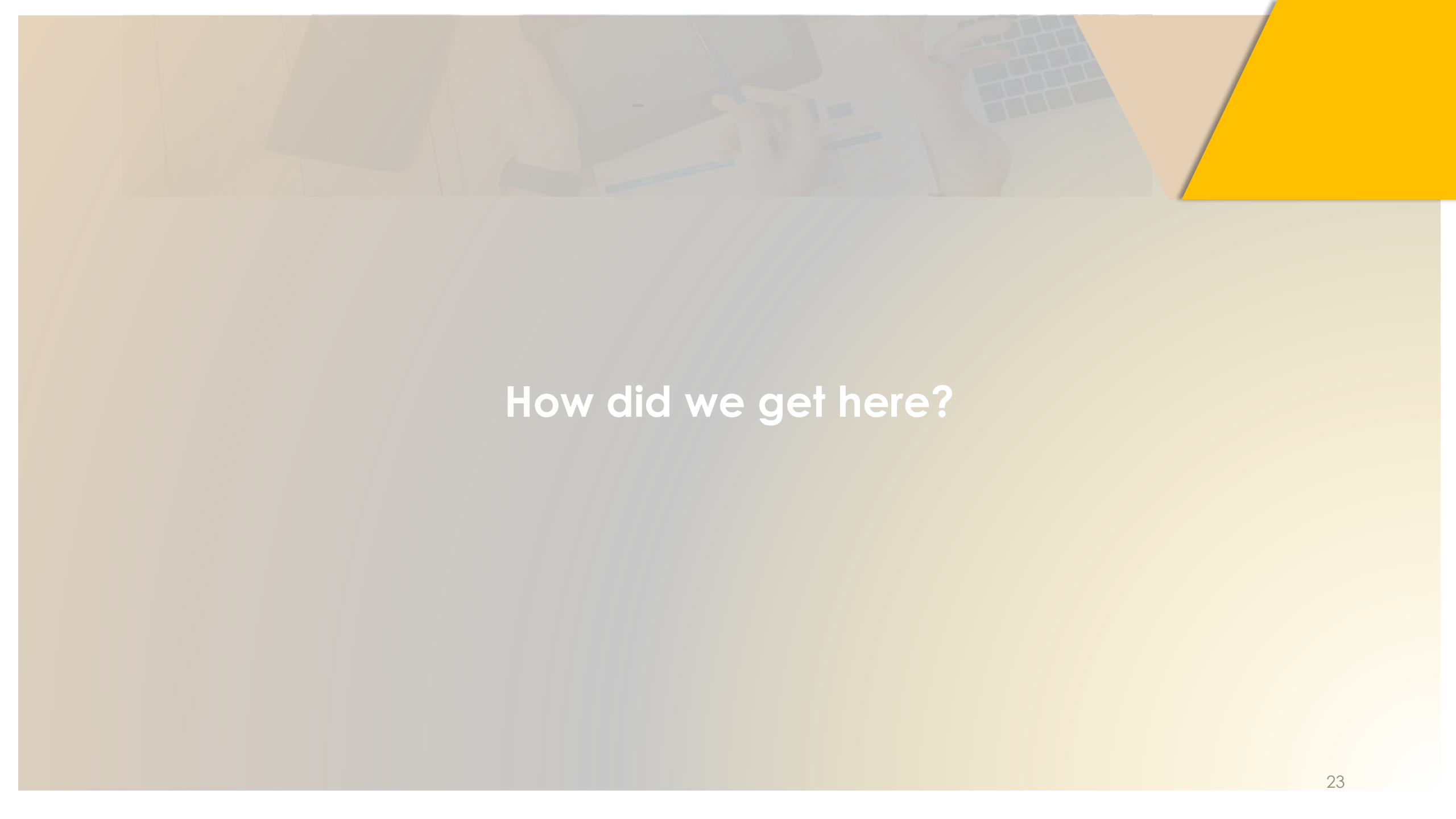
IA pour la recherche

V2X Semantic Anomaly Detection and Trust Reinforcement in C-ITS Communications

Détection de défaut d'arc électrique via une architecture multimodale arc

Hybrid Evaluation and Reconciliation of Query Results in Heterogeneous Data Lakes

Mitacs : Détection de biais dans les LLMs



How did we get here?

How did we get here?

+50 years!

AI HAS A LONG HISTORY OF BEING “THE NEXT BIG THING” ...



Timeline of AI Development

- **1950s-1960s:** First AI boom - the age of reasoning, prototype AI developed
- **1970s:** AI winter I
- **1980s-1990s:** Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
- **1990s:** AI winter II
- **1997:** Deep Blue beats Gary Kasparov
- **2006:** University of Toronto develops Deep Learning
- **2011:** IBM's Watson won Jeopardy
- **2016:** Go software based on Deep Learning beats world's champions

How did we get here?

1956 Dartmouth Conference: The Birth of Artificial Intelligence

Foundational Moment for AI: This event is often considered the **founding moment of AI** as a field of study. It was the first time researchers from various disciplines came together to explore the concept of machine intelligence.

AI HAS A LONG HISTORY OF BEING “THE NEXT BIG THING”...



Timeline of AI Development

- **1950s-1960s:** First AI boom - the age of reasoning, prototype AI developed
- **1970s:** AI winter I
- **1980s-1990s:** Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
- **1990s:** AI winter II
- **1997:** Deep Blue beats Gary Kasparov
- **2006:** University of Toronto develops Deep Learning
- **2011:** IBM's Watson won Jeopardy
- **2016:** Go software based on Deep Learning beats world's champions

How did we get here?

1958 Perceptron by Rosenblatt at Cornell

The perceptron, introduced by Frank Rosenblatt in 1958, is one of the earliest models of an artificial neuron, **laying the groundwork for the development of modern neural networks**. It was inspired by biological neurons and aimed to mimic the way the brain processes information.

FIG. 1 — Organization of a biological brain. (Red areas indicate active cells, responding to the letter X.)

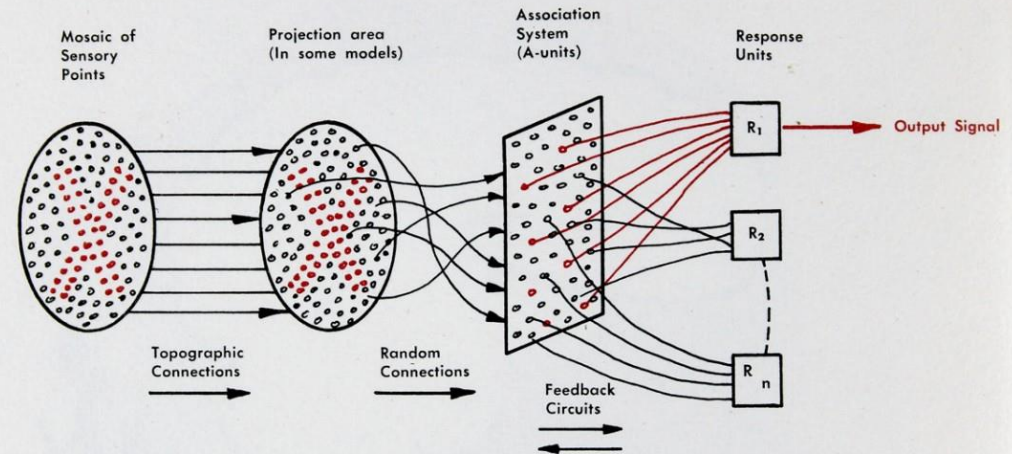


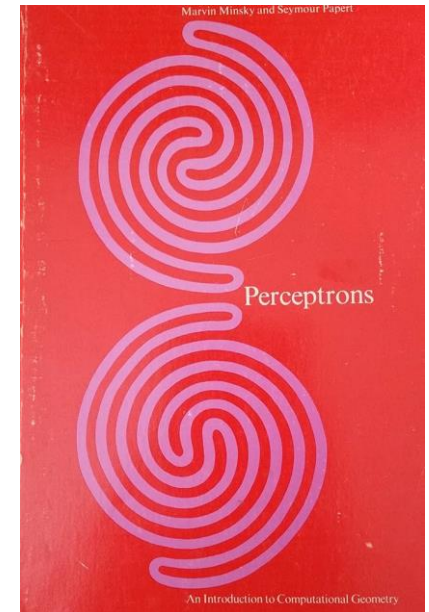
FIG. 2 — Organization of a perceptron.

How did we get here?

1969 Analysis of Perceptrons Minsky & Papert

"Perceptrons: An Introduction to Computational Geometry," Marvin Minsky and Seymour Papert **critically analyzed the capabilities** and limitations of perceptrons

Their work highlighted fundamental challenges in the perceptron model, particularly **its inability to solve problems that are not linearly** separable, such as the XOR problem.



How did we get here?

1st AI Winter: 1974 - 1980

An AI Winter refers to a period of **reduced funding**, interest, and activity in artificial intelligence research.

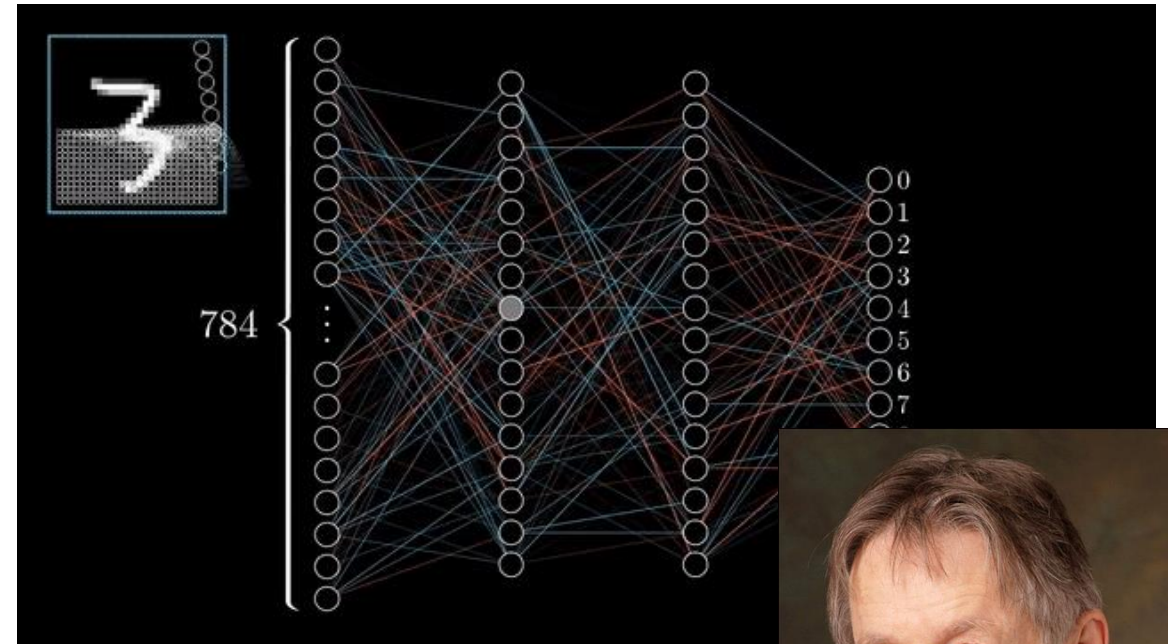
During these times, enthusiasm for AI wanes due to unmet expectations, and progress in the field slows down significantly.



How did we get here?

Backpropagation – Popularized in 1986 by Hinton et.al

Backpropagation, or “*backward propagation of errors*,” is an algorithm used to train neural networks by efficiently computing the gradient of the loss function with respect to the network’s weights. This process allows the network to adjust its weights to minimize errors, thereby improving its performance.



Geoffery Hinton

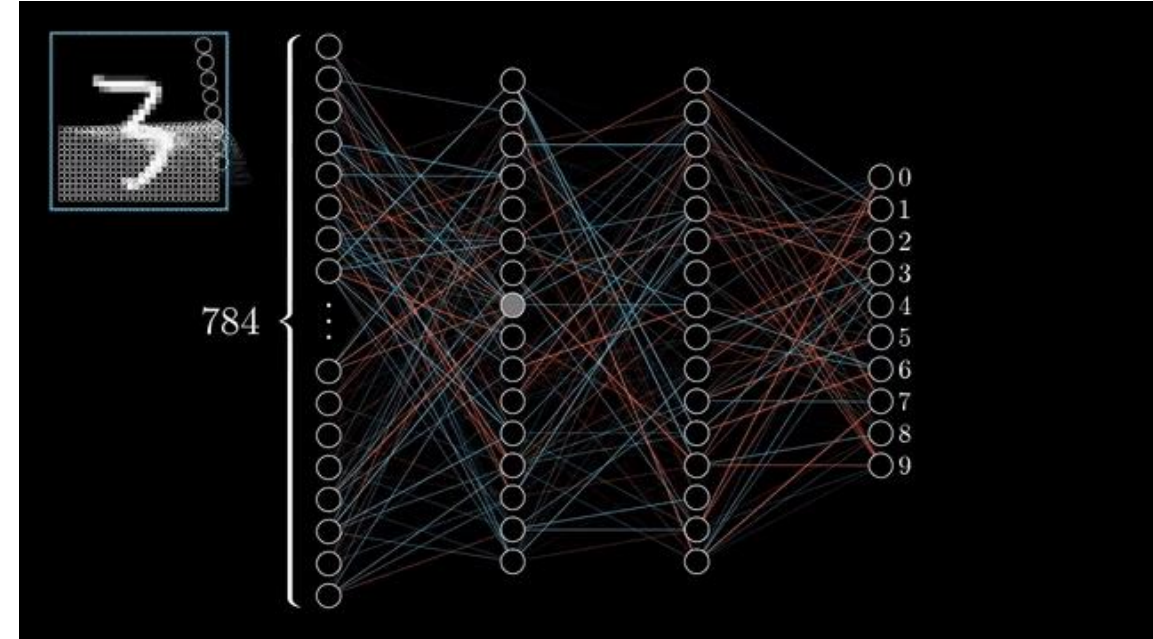
Nobel prize in Physics (2024)
Turing Award (2019)



How did we get here?

Backpropagation –
Popularized in 1986 by Hinton
et.al

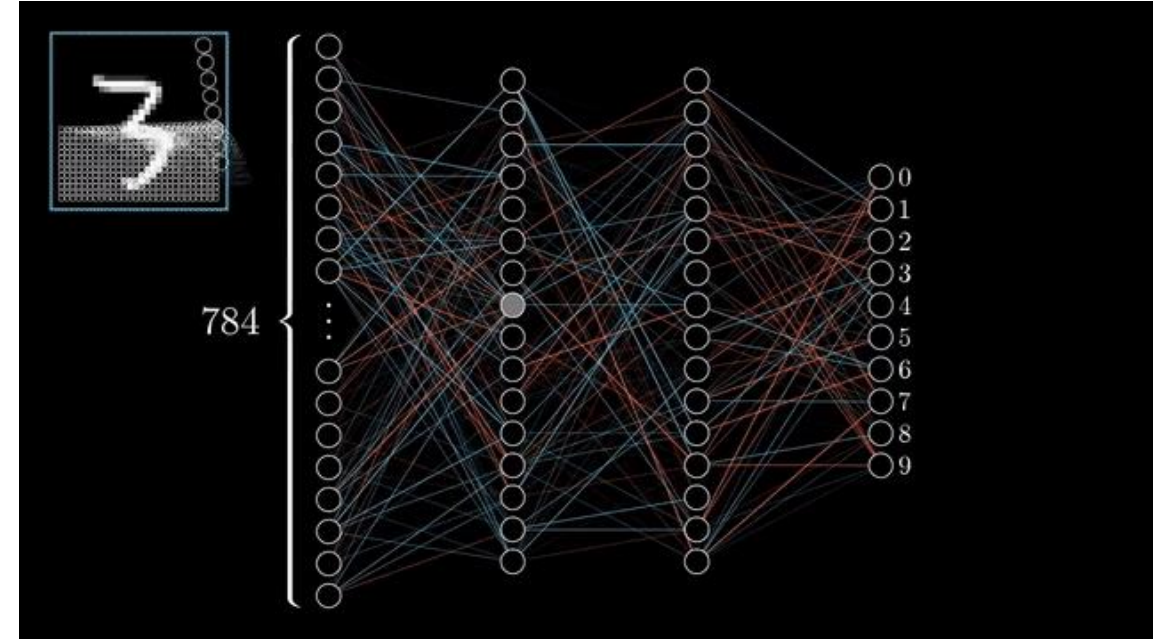
- It uses **the chain rule** from calculus to compute gradients layer by layer, starting from the output layer and moving backward through the network.
- Before backpropagation, neural networks were primarily limited **to linear models with simple learning capabilities**.



How did we get here?

Backpropagation – Popularized in 1986 by Hinton et.al

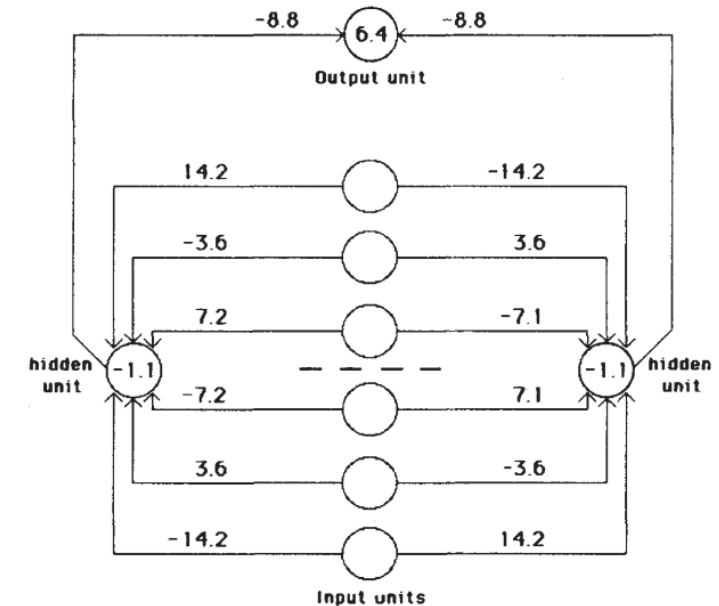
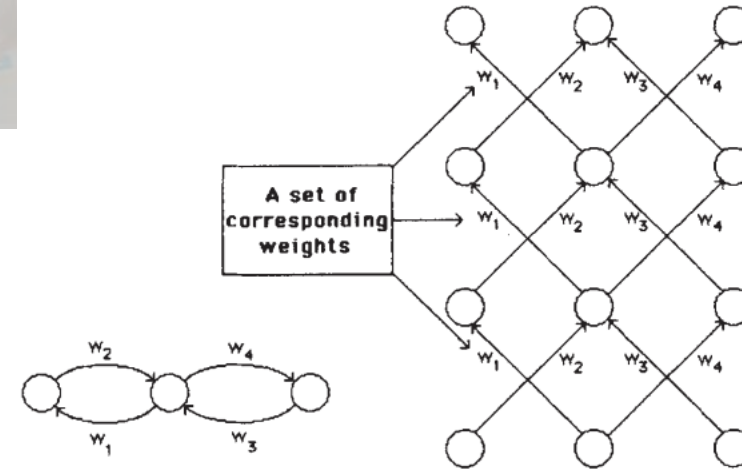
- The introduction of backpropagation demonstrated that neural networks could approximate any continuous function, given sufficient data and appropriate network architecture.



How did we get here?

1986 - Multi-layer Perceptron

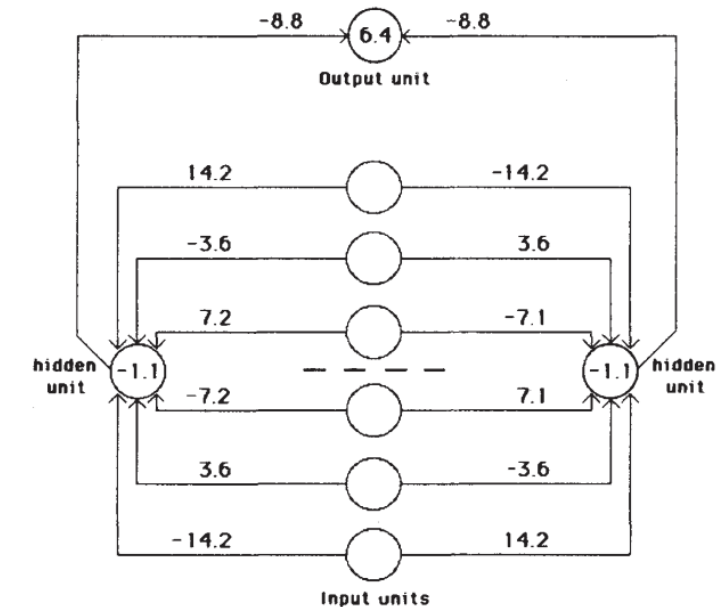
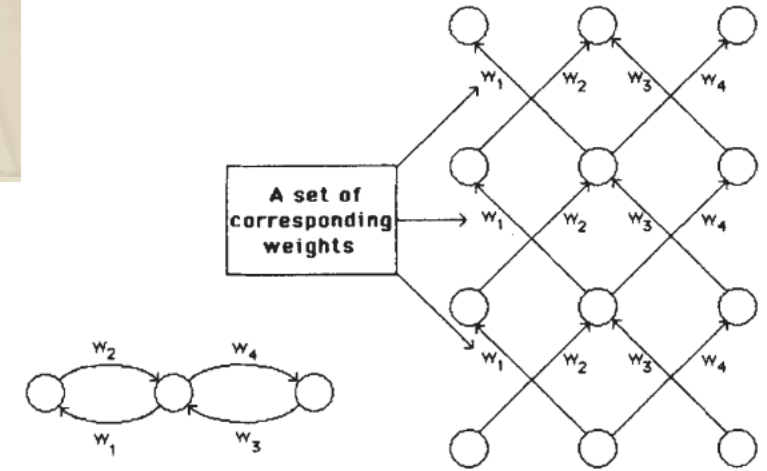
Learning Internal Representations: The authors demonstrated that MLPs could learn internal representations **in hidden layers**, which are crucial for capturing complex features in data. This ability to learn hierarchical representations was a significant advancement over previous models.



How did we get here?

1986 - Multi-layer Perceptron

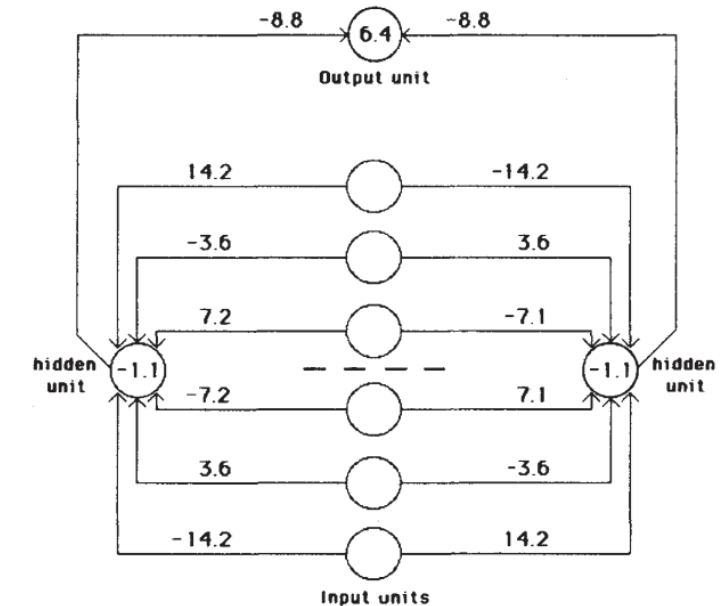
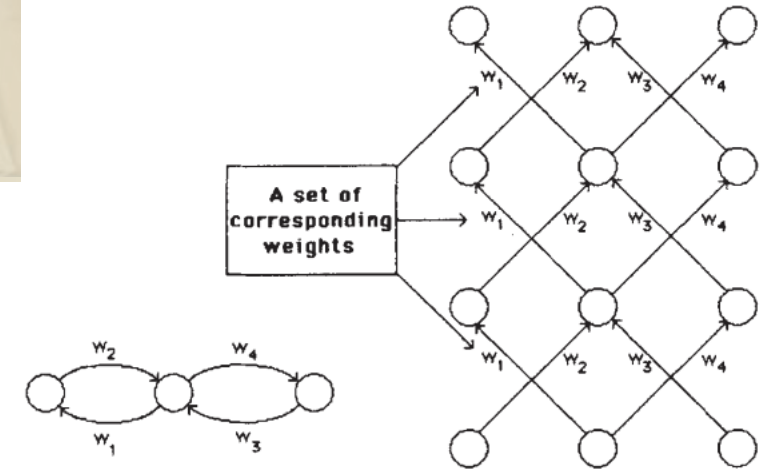
Non-Linear Capabilities: The paper highlighted that **MLPs with non-linear activation functions** in the hidden layers could solve problems that linear models could not, such as XOR, which was a known limitation of single-layer perceptrons.



How did we get here?

1986 - Multi-layer Perceptron

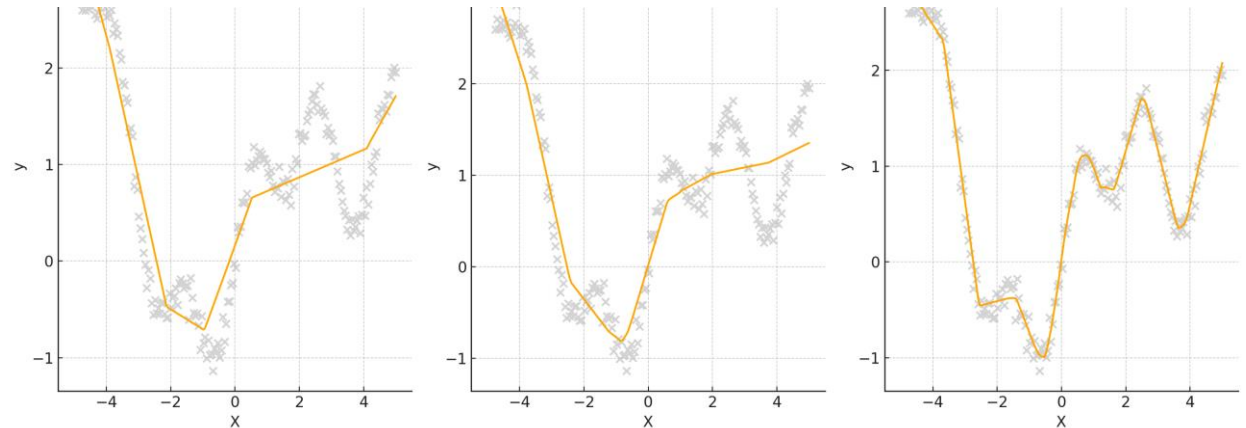
Empirical Validation: The 1986 paper provided empirical results showing that MLPs, trained using backpropagation, could successfully perform tasks like pattern recognition, proving the effectiveness of the approach in practical applications.



How did we get here?

1989 – Universal Function Approximation Theory

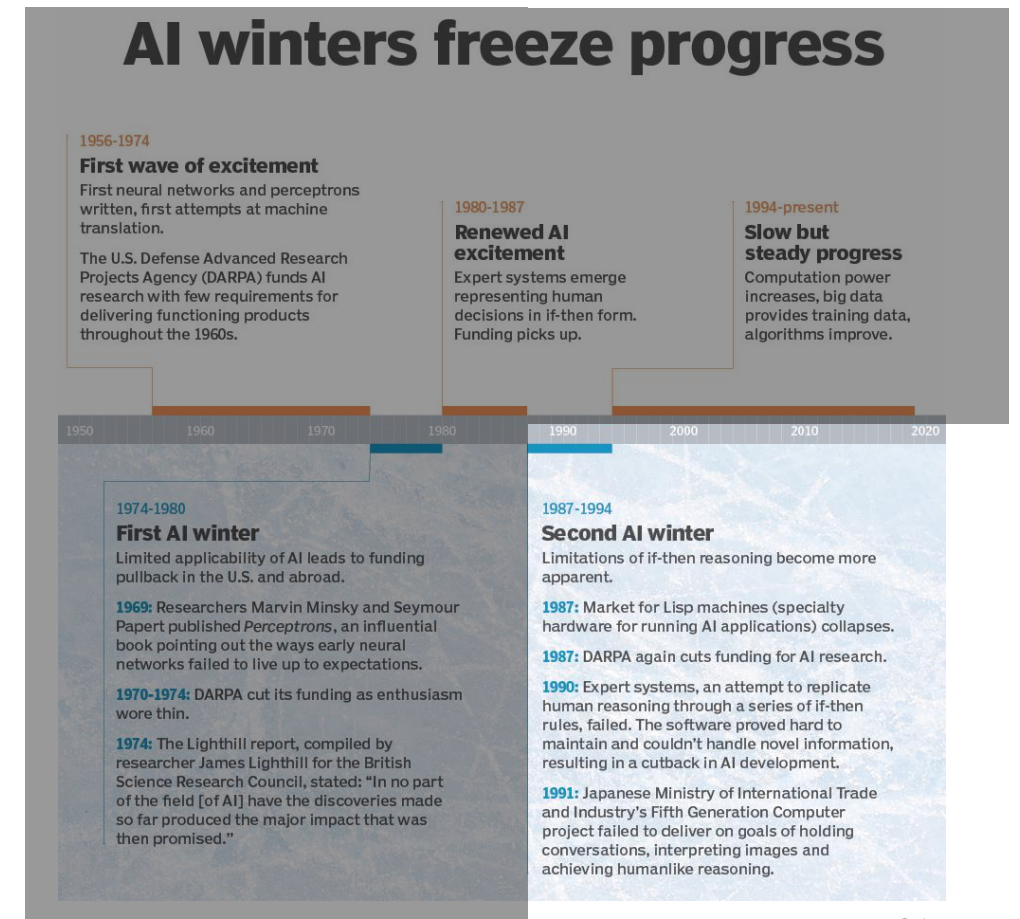
The Universal Approximation Theorem, proposed by George Cybenko in 1989 states that a feedforward neural network with a single hidden layer, containing a finite number of neurons and using a non-linear activation function (like the sigmoid function), can approximate any continuous function, given sufficient neurons in the hidden layer.



How did we get here?

What led to the second AI winter?

1. **Unmet Expectations:** Expert systems, initially promising, failed to generalize beyond narrow domains, leading to disillusionment.
2. **Economic Recession:** The late 1980s recession led to budget cuts in AI research, shifting focus to more immediate technologies.
3. **Technological Limits:** Insufficient computational power and algorithmic challenges hindered AI development.
4. **Reduced Interest:** Funding and interest in AI dropped, leading to a slowdown in research and fewer innovations.
5. **Legacy and Recovery:** Lessons from this period set the stage for the AI resurgence in the late 1990s, paving the way for modern AI advancements.



How did we get here?

2012 AlexNet – Computer Vision and GPUs Dominate AI

Revolutionary Performance:

AlexNet, developed by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, won the 2012 ImageNet competition with a large margin.

GPU Acceleration:

AlexNet utilized GPUs (specifically NVIDIA GPUs) to dramatically reduce the training time for deep neural networks. This approach allowed the model to be trained on a much larger scale.

Catalyst for AI Growth:

The success of AlexNet spurred widespread adoption of deep learning and GPUs in AI research and industry.

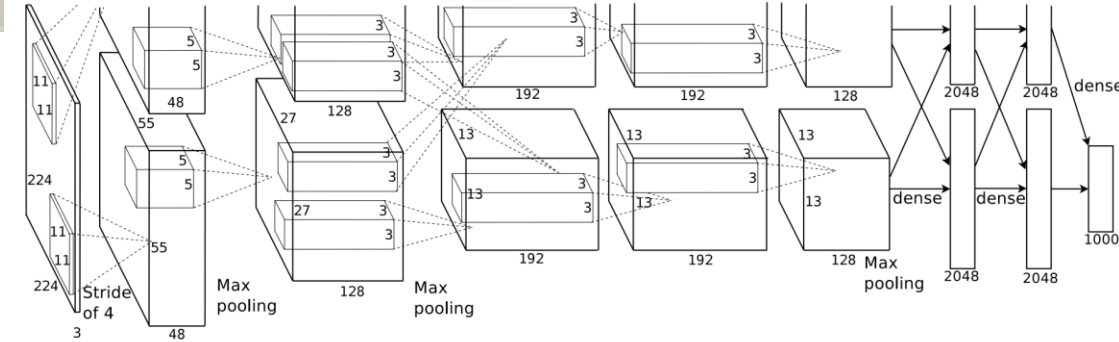
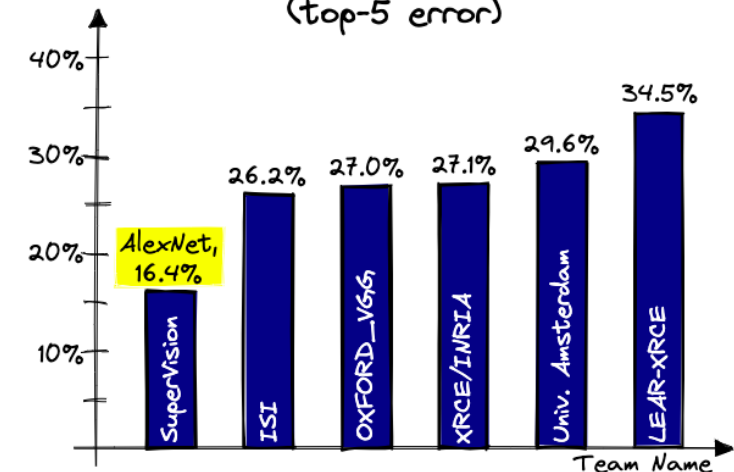


Figure 2: An illustration of the architecture of our CNN, explicitly showing the delineation of responsibilities between the two GPUs. One GPU runs the layer-parts at the top of the figure while the other runs the layer-parts at the bottom. The GPUs communicate only at certain layers. The network's input is 150,528-dimensional, and the number of neurons in the network's remaining layers is given by 253,440–186,624–64,896–64,896–43,264–4096–4096–1000.

2012 ImageNet Challenge
(top-5 error)



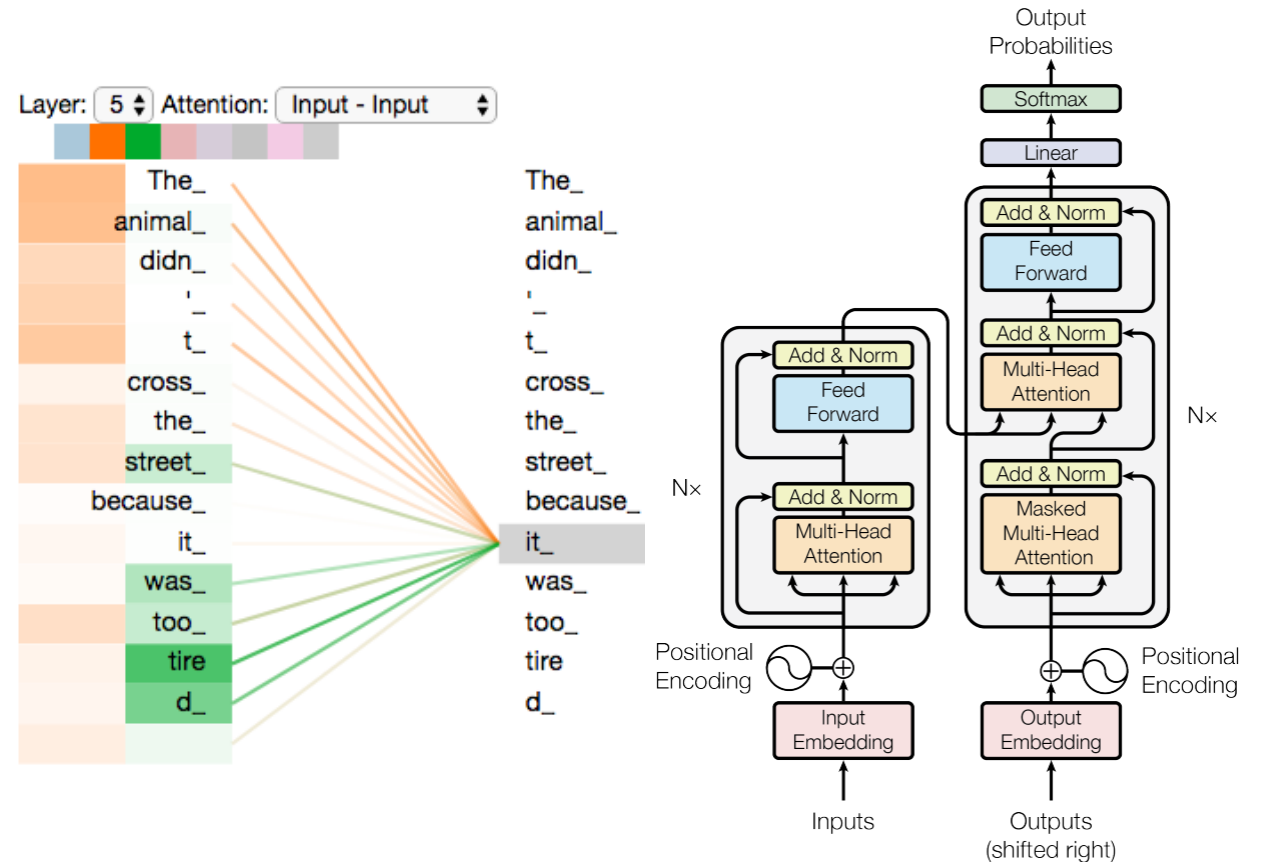
How did we get here?

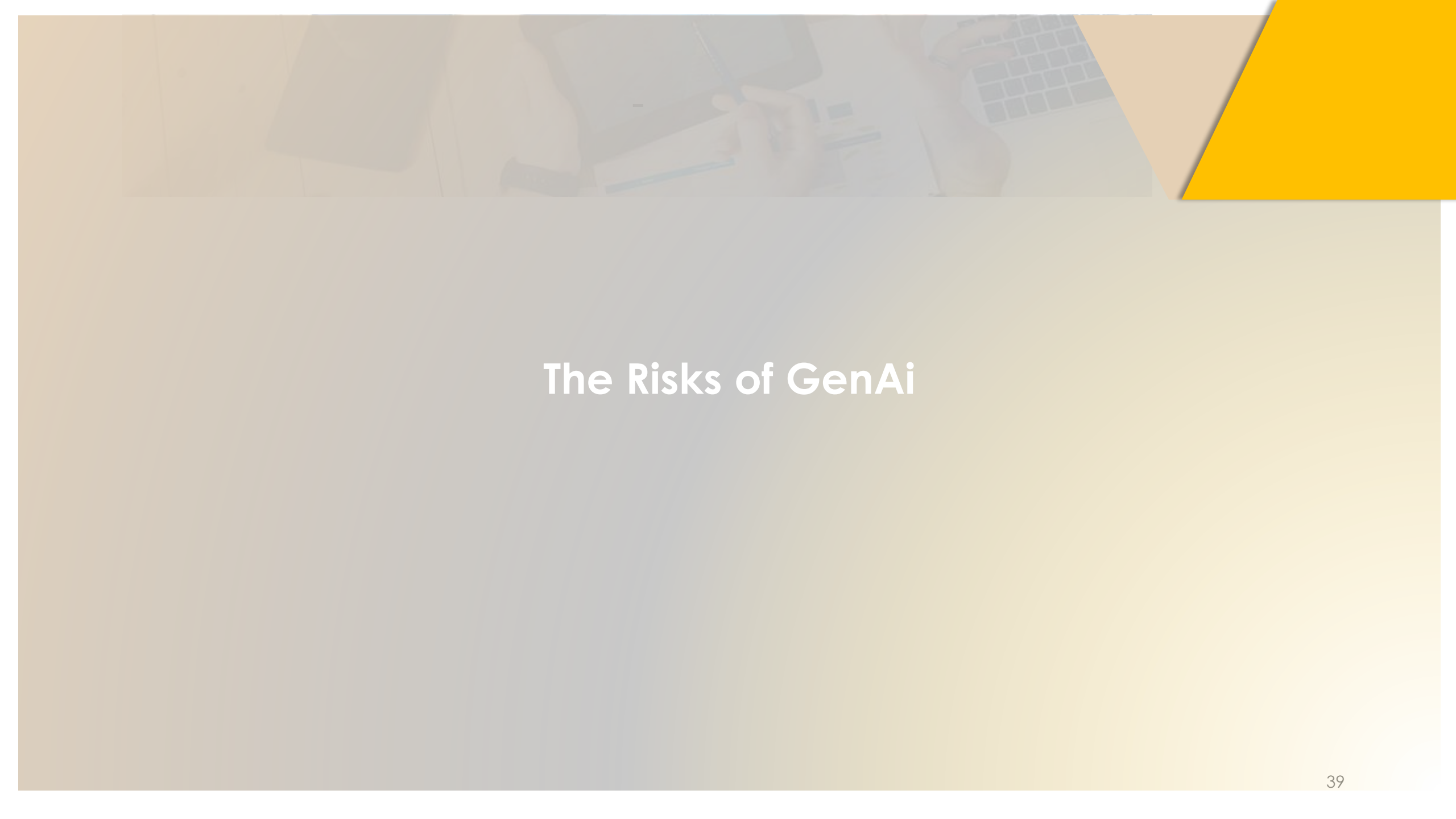
2017++: Attention is all you Need – Everything Changes

In the 2017 paper, Attention is all you need, a new deep learning architecture, the Transformer was introduced.

This model, and its variants: BERT and GPT have been found to solve traditional natural language problems and are capable of generating complex semantic outputs

These text generation models, together with the image generation models constitute Generative AI and will be the focus moving forward.



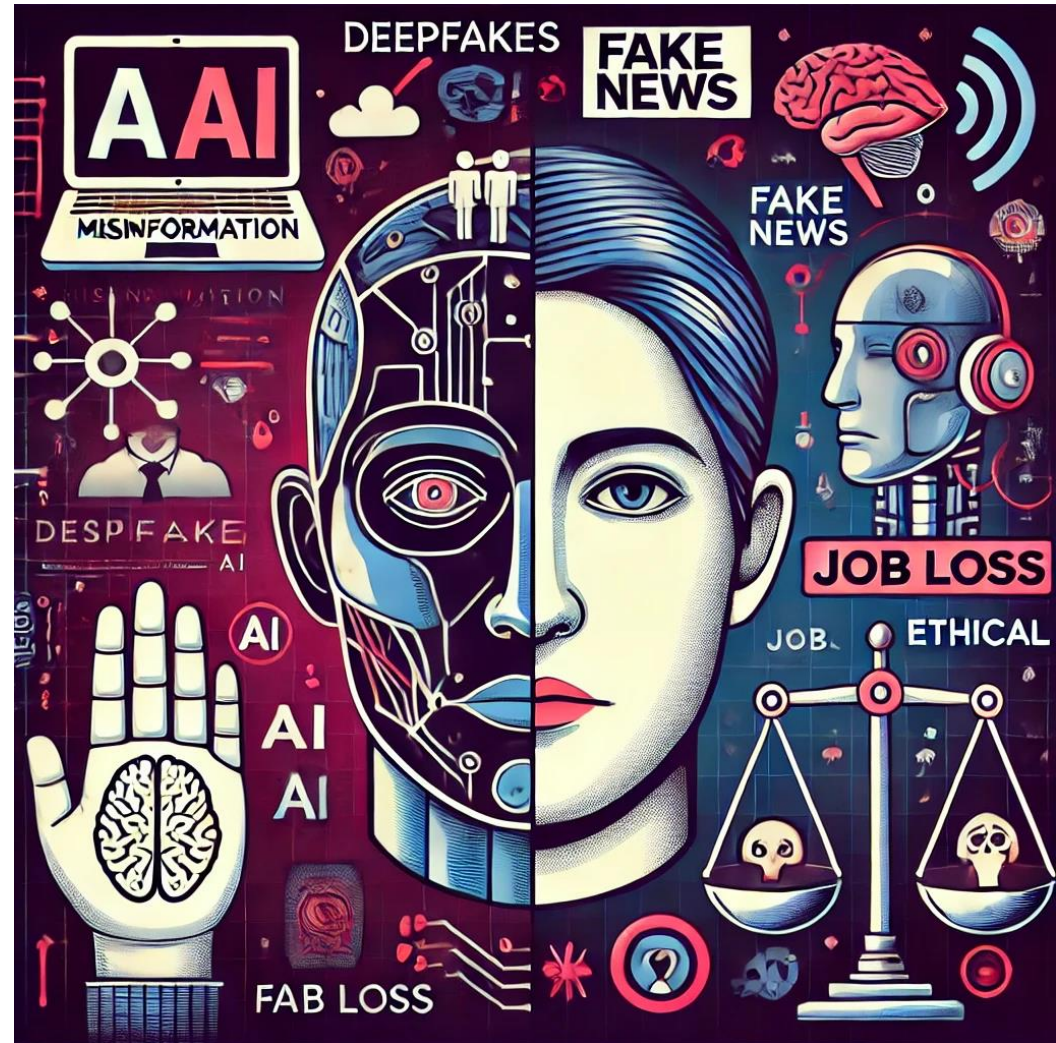


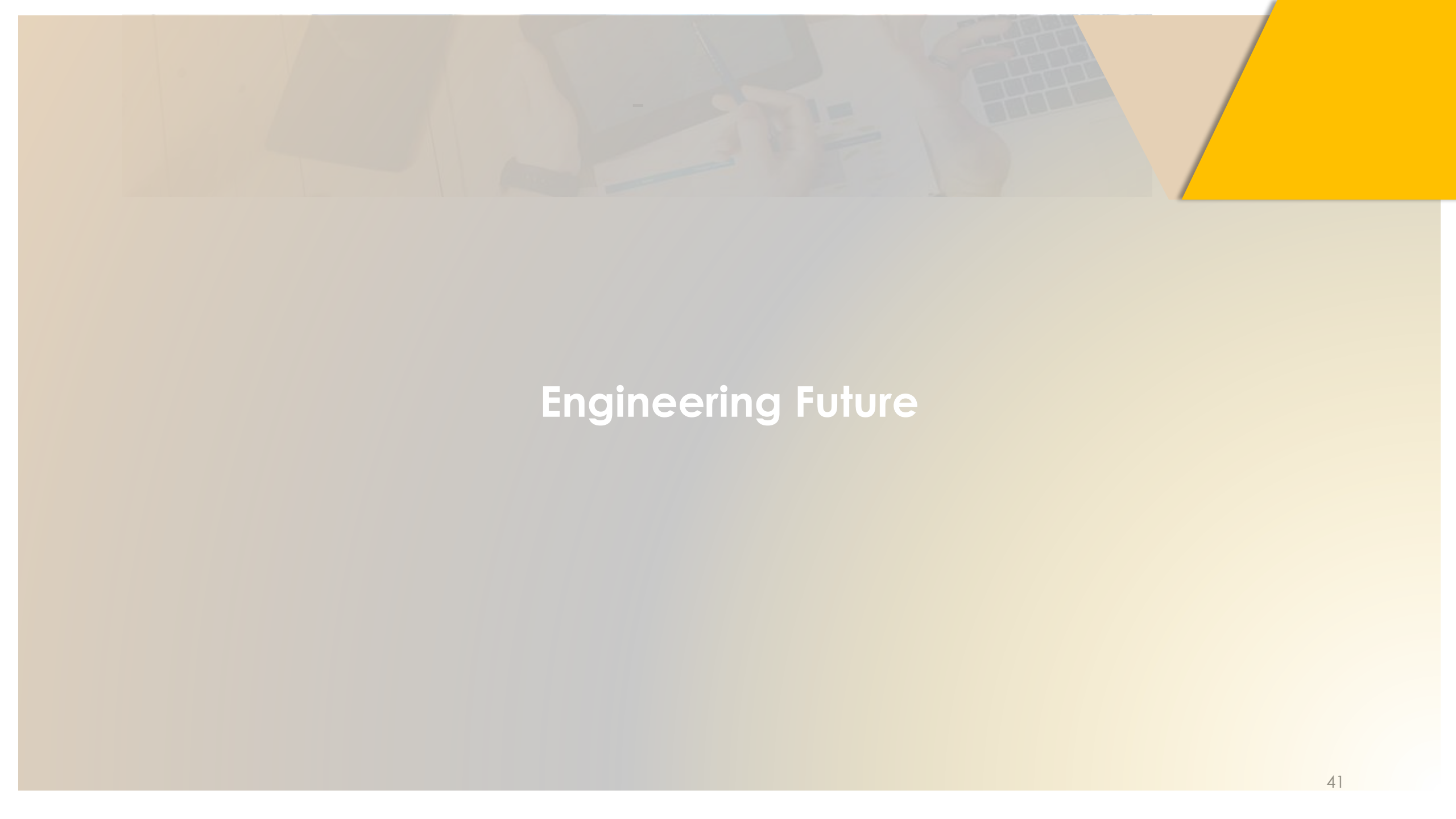
The Risks of GenAi

Risks

For all its technological wonder, GenAI poses a serious challenge some fundamental parts of modern society:

- Trust and Information
- Deepfake
- Digital records of events and people
- Jobs
- Humanity extinction





Engineering Future

Risks

Integrating LLMs (Large Language Models) **everywhere**, even where a single line of traditional code is enough. It highlights the absurdity of using a massive and expensive server infrastructure for a simple capitalization check.

coding in 2022

```
JS stringUtils.js ×  
1 function startsWithCapitalLetter(str) {  
2   return str.length > 0 && str[0] === str[0].toUpperCase();  
3 }
```

coding in 2027

```
JS stringUtils.js ×  
1 import Anthropic from '@anthropic-ai/sdk';  
2  
3 const anthropic = new Anthropic({  
4   apiKey: process.env.ANTHROPIC_API_KEY,  
5 });  
6  
7 export async function startsWithCapitalLetter(str) {  
8   const prompt = `You are a capitalization expert.  
9     You have been studying capitalization for years.  
10  
11     Text to analyze: "${str}"  
12  
13     Respond with ONLY 'true' or 'false'.  
14     Is the first letter of the text a capital letter?`;  
15  
16   const response = await anthropic.messages.create({  
17     model: 'claude-mythos-1-20261225',  
18     max_tokens: 5000000,  
19     temperature: 0,  
20     thinking: 'ultramaxmegathink',  
21     messages: [{ role: 'user', content: prompt }],  
22   });  
23  
24   const answer = response.content[0].text.trim().toLowerCase();  
25   return answer === 'true';  
}
```

Knowledge → Synthesis → Problem Solving → Demonstrable Value



The Reality of the AI Job Market: A Correction Phase



2021

The Slowdown

The pandemic paralyzes hiring and creates an industrial bottleneck.



2022 – 2023

The Frenetic Hiring Boom

In the rush to adopt AI, companies are hiring fast and investing heavily — often without a clear strategy.



2024 – 2025

The Great Awakening

The easy money is running out. Companies now require real skills, proven experience, profitability and business value.



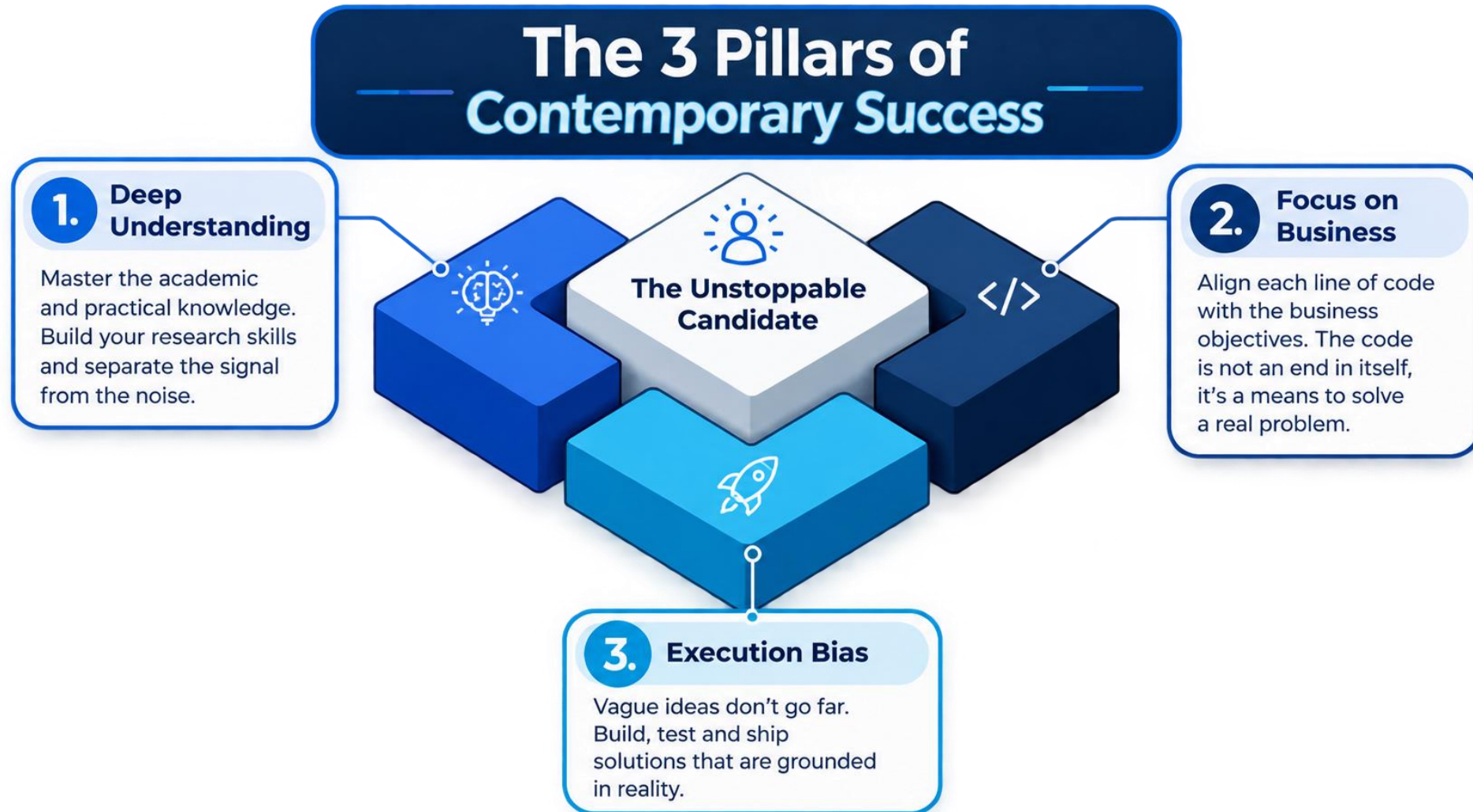
- ✓ Real skills
- ✓ Proven experience
- ✓ Profitability
- ✓ Business value



The market is not dead. The opportunity remains massive,
but it now requires a strategic approach, not just a single CV.



Knowledge → Synthesis → Problem Solving → Demonstrable Value



Knowledge → Synthesis → Problem Solving → Demonstrable Value

⚡ The Coding Vibe and the Technical Debt ⚡



AI generates code instantly, but the engineer remains the guardian of technical debt.



The Good Debt (The Real Estate of the Future)

- ✓ Clear and well-defined objectives
- ✓ Immediate value delivered
- ✓ Architecture designed and documented by humans
- ✓ Returns on investment outweigh the maintenance cost



The Bad Debt (The Credit Card)

- ✗ Code generation is impulsive and unplanned
- ✗ Spaghetti code and outdated APIs
- ✗ Solutions in a rush to fix problems
- ✗ Maintenance and debugging become impossible

Knowledge → Synthesis → Problem Solving →
Demonstrable Value

> The 3 Steps to Succeed in AI Development <

1



Diversify Your Skill Set

Don't be a one-trick pony. Master scaling, the user experience (UX) and the classic software architecture alongside LLMs.

2



Develop the Tech Guardian

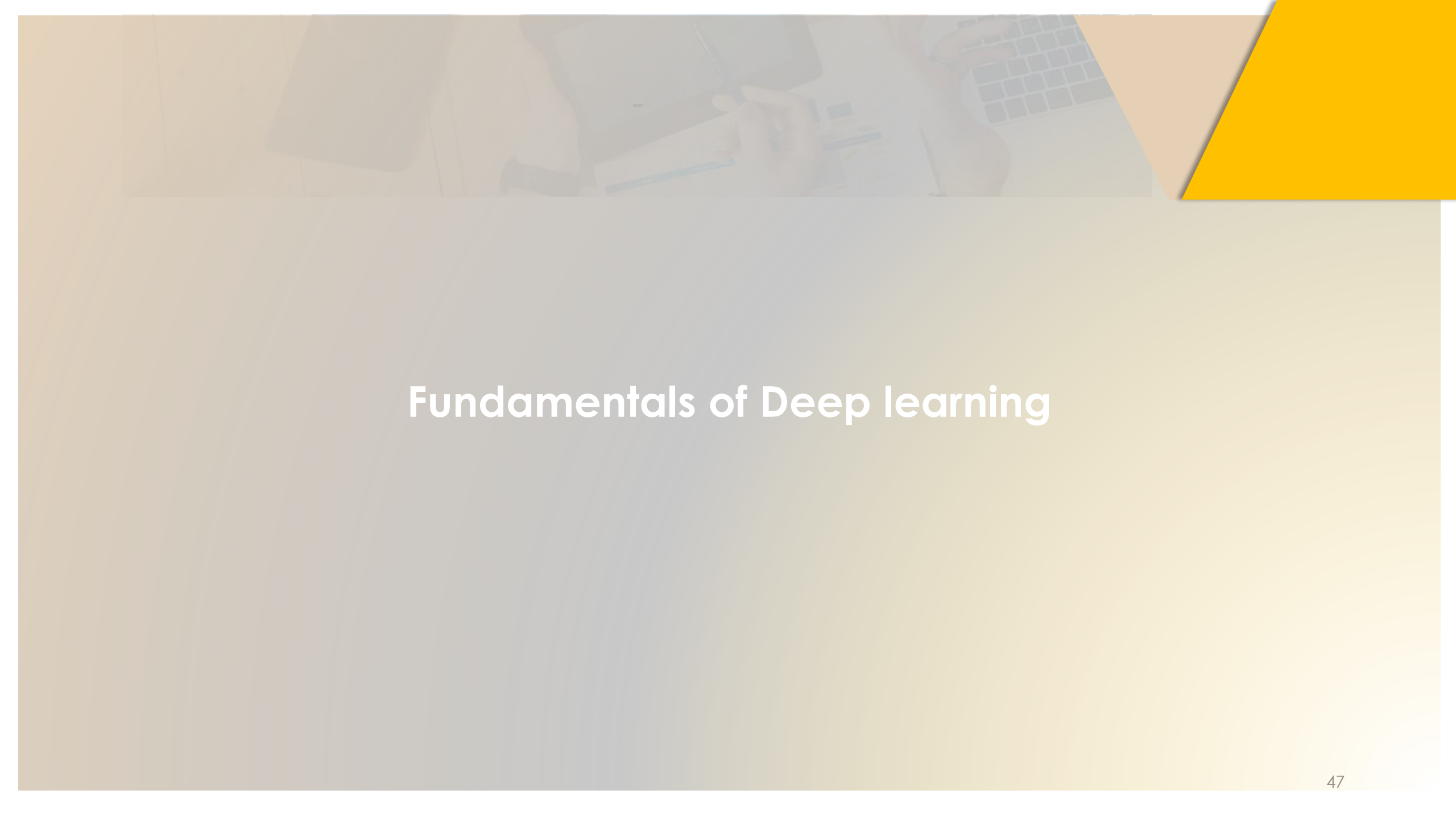
Use AI to code for life, but mostly, keep the human understanding and system maintenance. Align each line with a real business need.

3



Experiment Without Limits

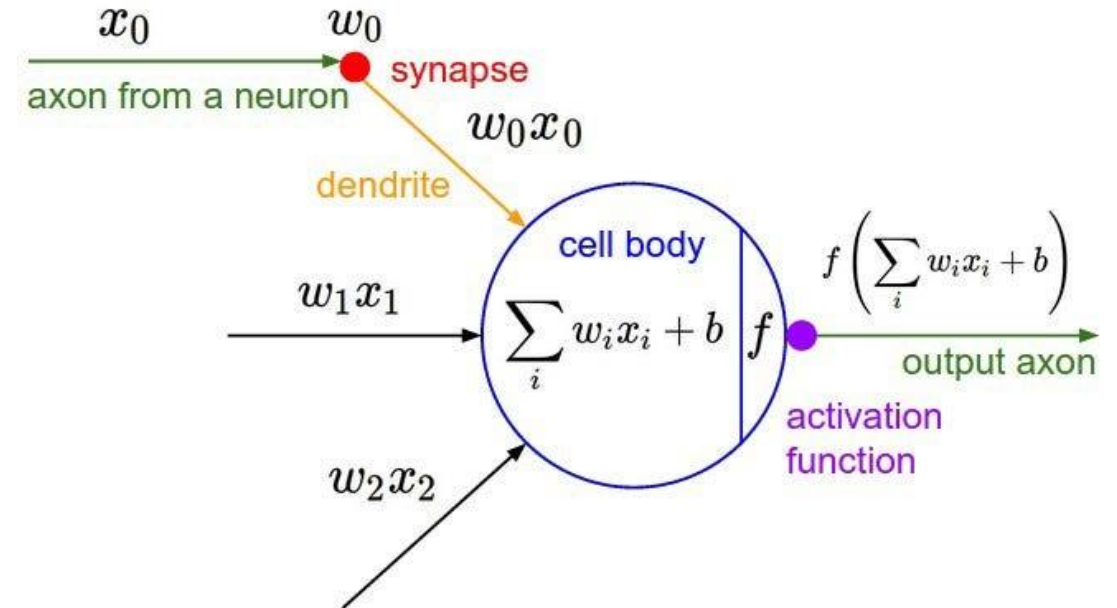
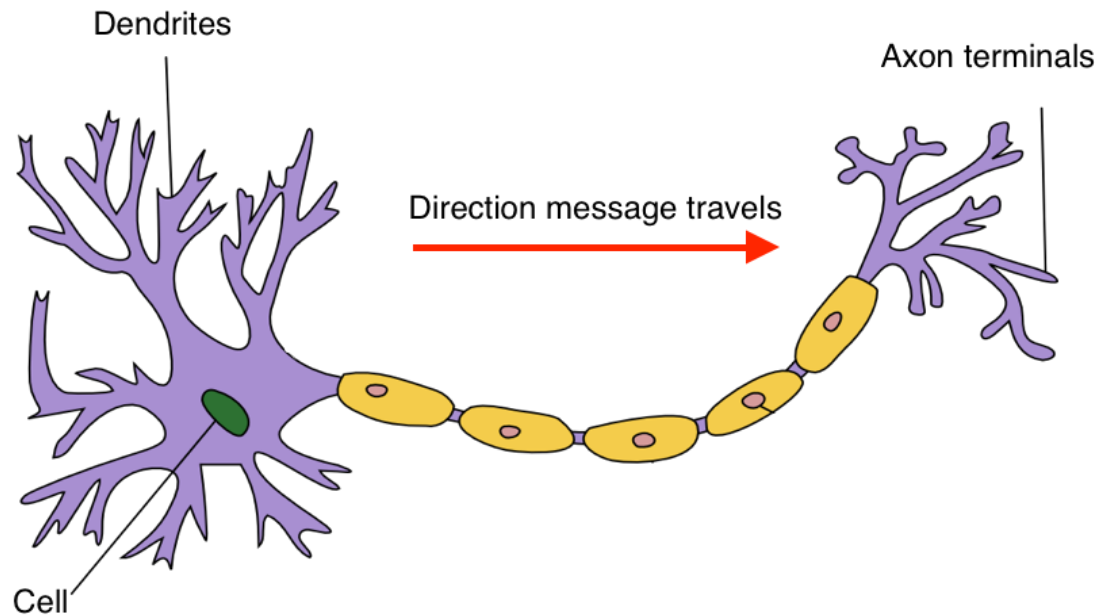
The cost of failure has never been lower. Launch weekend projects; a failure isn't a cost, it's an invaluable learning experience.



Fundamentals of Deep learning

FDL : The neuron

This is called a neural network. Each neuron would try to linearly represent the input data.



FDL : loss curve

The gradient tells us the direction. The optimizer takes the steps.

Chain Rule

To minimize the loss, we compute the gradient of the loss with respect to the model parameters θ using the chain rule:

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z} \cdot \frac{\partial z}{\partial \theta}$$

Simple example

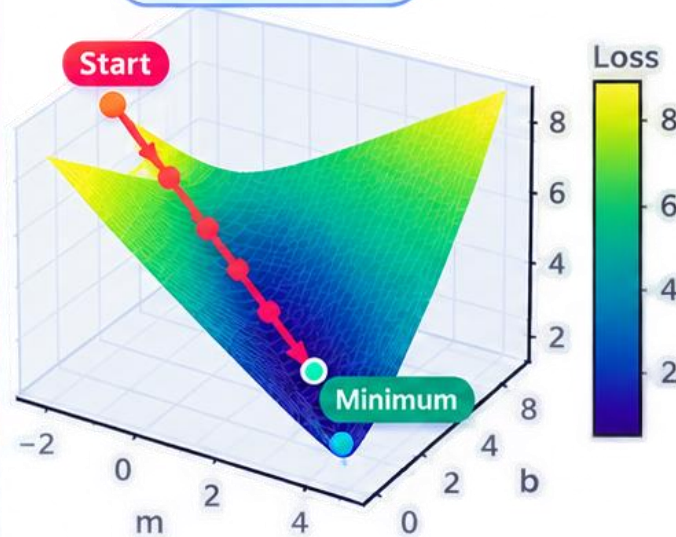
For a linear model: $\hat{y} = wx + b$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w} = (\hat{y} - y) \cdot x$$



This gradient tells us how to update w (and b) to reduce the loss.

Loss Surface



Filled Contours Plot



The Optimization Process

- 1 Compute the gradient using the chain rule
- 2 Update the parameters (e.g. $\theta \leftarrow \theta - \eta \nabla L$)
- 3 Repeat until convergence (minimum loss)



There are **many optimizers** that do this!

Gradient Descent, Adam, RMSprop, Adagrad, and others... Each uses a different strategy to follow the gradient and find the minimum.

Submission 1 : optimization algorithms

Generate in JSON Format, an explication of those three optimizers : Vanilla Gradient, SGD, AdamW.

Criterion	Code	Weight	Evaluation criteria
Relevance	R	0.20	Measures how well the response addresses the question, follows the requested scope, and focuses on information that is directly useful to the user.
Pedagogical quality	P	0.25	Evaluates the clarity, organization, accessibility, and explanatory value of the response, including the use of appropriate examples or explanations.
Technical accuracy	A	0.30	Assesses whether the technical information is correct, precise, consistent, and free from significant conceptual or factual errors.
Comparison and critical thinking	C	0.15	Measures the quality of comparisons, identification of similarities and differences, consideration of trade-offs, and depth of critical analysis.
Support quality	S	0.05	Evaluates whether claims are adequately supported by explanations, evidence, examples, or appropriate justification.
Sources and reproducibility	X	0.05	Assesses the quality, transparency, traceability, and reproducibility of the sources or methods used.

Final score

$$\text{Final score} = 0.20R + 0.25P + 0.30A + 0.15C + 0.05S + 0.05X$$

CNN : MCQ

Question 1

What is a Convolutional Neural Network mainly designed to process?

- A. Only numerical tables
- B. Grid-like data such as images
- C. Audio signals exclusively
- D. Database queries

Correct answer: B

Explanation: CNNs are designed to process grid-like data, especially images.

CNN : MCQ

Question 2

What is the main role of convolutional layers in a CNN?

- A. To store the training dataset
- B. To detect local patterns such as edges, textures, and shapes
- C. To randomly modify the input image
- D. To convert images into text automatically

Correct answer: B

Explanation: Convolutional filters detect local visual features, such as edges, textures, and shapes.

CNN : MCQ

Question 3

Which type of feature can a convolutional filter detect?

- A. Edges
- B. Textures
- C. Shapes
- D. All of the above

Correct answer: D

Explanation: Convolutional layers can learn to detect different visual patterns, including edges, textures, and shapes.

CNN : MCQ

Question 4

What is the purpose of an activation function such as ReLU?

- A. To introduce nonlinearity into the network
- B. To reduce the size of the image
- C. To store feature maps
- D. To remove all features from the image

Correct answer: A

Explanation: Activation functions such as ReLU introduce nonlinearity, allowing the network to learn complex relationships.

CNN : MCQ

Question 5

What is the main purpose of pooling layers?

- A. To increase the spatial dimensions of feature maps
- B. To reduce the spatial dimensions of feature maps
- C. To replace the convolutional layers
- D. To generate new training labels

Correct answer: B

Explanation: Pooling reduces the spatial dimensions of feature maps, which improves computational efficiency.

CNN : MCQ

Question 6

How can pooling make a CNN more robust?

- A. By making it less sensitive to small translations in the input
- B. By increasing the number of pixels
- C. By removing the need for training
- D. By converting the model into a linear model

Correct answer: A

Explanation: Pooling helps the model remain relatively robust when features move slightly within the image.

CNN : MCQ

Question 7

What is the role of fully connected layers in a CNN?

- A. To detect only edges
- B. To reduce image resolution
- C. To integrate extracted features and perform classification
- D. To apply filters to local image regions

Correct answer: C

Explanation: Fully connected layers combine the features extracted by previous layers and use them to classify the input.

CNN : MCQ

Question 8

What does end-to-end learning mean in a CNN?

- A. The model learns features directly from raw data
- B. The model requires manually designed features
- C. The model only learns the final classification layer
- D. The model cannot process images

Correct answer: A

Explanation: CNNs can learn useful features directly from raw input data without requiring manual feature extraction.

CNN : MCQ

Question 9

Which sequence best represents the typical processing flow of a CNN?

- A. Fully connected layers → pooling → convolution → input
- B. Input → convolution → activation → pooling → fully connected layers
- C. Pooling → input → fully connected layers → convolution
- D. Activation → input → classification → convolution

Correct answer: B

Explanation: A typical CNN applies convolutional layers, activation functions, and pooling before using fully connected layers for classification.

CNN : MCQ

Question 10

What is a major advantage of CNNs for image classification?

- A. They require manual extraction of every visual feature
- B. They can automatically learn hierarchical visual features
- C. They work only with black-and-white images
- D. They do not require training data

Correct answer: B

Explanation: CNNs automatically learn features ranging from simple patterns, such as edges, to more complex structures, such as shapes and objects.

FDL : Data augmentation

In PyTorch, the **torchvision.transforms** module is used to prepare images before sending them into the neural network.

With it, you can:

- Convert an image to a tensor (ToTensor()),
- Resize it (Resize()),
- Normalize it (Normalize()),
- Apply random transformations (rotation, cropping, etc.) to augment the data.

```
random_transforms = transforms.Compose([
    transforms.RandomRotation(5),
    transforms.RandomResizedCrop((IMG_WIDTH, IMG_HEIGHT), scale=(.9, 1), ratio=(1, 1)),
    transforms.RandomHorizontalFlip(),
    transforms.ColorJitter(brightness=.2, contrast=.5)
])
```

`transforms.Compose()` allows you to chain multiple transformations together in a pipeline.

FDL : Transfer Learning

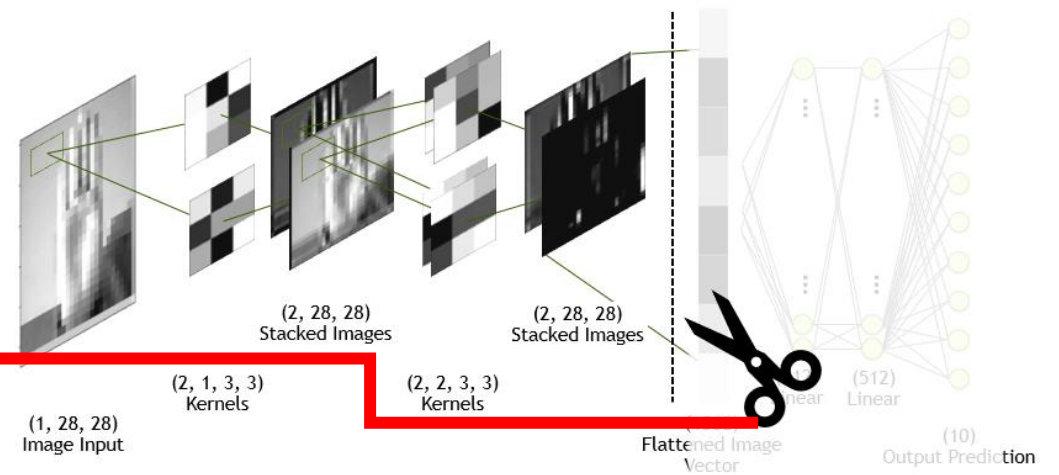
The VGG16 model is composed of two main parts:

A **feature extractor** → features

A **classifier (fully connected head)** → classifier

Avgpool to set the size of image to fed to the classifier

```
VGG(  
  (features): Sequential( ... 31 layers ... )  
  (avgpool): AdaptiveAvgPool2d(output_size=(7, 7))  
  (classifier): Sequential(  
    (0): Linear(in_features=25088, out_features=4096, bias=True)  
    (1): ReLU(inplace=True)  
    (2): Dropout(p=0.5, inplace=False)  
    (3): Linear(in_features=4096, out_features=4096, bias=True)  
    (4): ReLU(inplace=True)  
    (5): Dropout(p=0.5, inplace=False)  
    (6): Linear(in_features=4096, out_features=1000, bias=True)  
  )  
)
```



Hyperparameters, Early Stops!

Hyperparameters are **configuration values** that are not learned **automatically** from the data (unlike the model's weights and biases).

Optimization Examples:

- **learning_rate, optimizer (Adam, SGD, etc.),**
- Description: Control how the model updates its weights during training.

Architecture Examples:

- **number_of_layers, number_of_neurons, kernel_size, activation_function**
- Description: Define the structure of the neural network.

Regularization Examples:

- **dropout_rate, batch_norm,**
- Description: Prevent overfitting and improve generalization.

Training process Examples:

- **batch_size, epochs, shuffle, early_stopping**
- Description: Control how the model sees and learns from data.

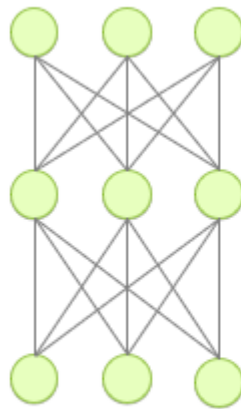
Data-related Examples:

image_size, augmentation parameters, normalization

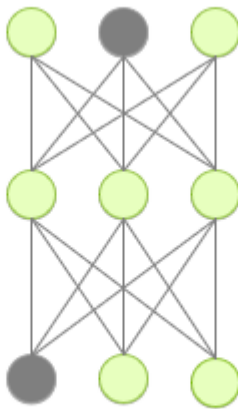
Description: Affect the preprocessing and input data format.

FDL : Regularization with dropout

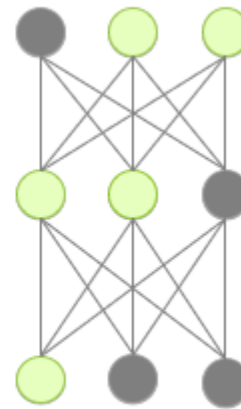
Dropout is another interesting tool which in a way, reflects ensemble learning. During training, we're going to randomly shutoff neurons at a rate we specify, meaning, it's unable to learn for that step of training. This prevents overfitting



rate = 0



rate = .2



rate = .4

FDL : Deployment

You can save your model (Pytorch version)

```
torch.save(model.state_dict(), "model.pth")
```

state_dict() will contain these informations.
(You can save the full model)

```
conv.0.weight  
conv.0.bias  
conv.3.weight  
conv.3.bias  
fc.1.weight  
fc.1.bias
```

Recreate your architecture in the API (REST for example) and then load the parameter

```
model = MyCNN()                # same architecture  
model.load_state_dict(torch.load("model.pth"))  
model.eval()                   # set to evaluation mode
```

Submission 2 : CNN challenge

Go to :
mariemzaouali.github.io and
click on **Enseignement>GenAI**

mariemzaouali.github.io/2026-09-24-genAItp1/

Adobe Acrobat

ENSEIGNEMENT RECHERCHE A PROPOS DE MOI Télécharger PDF

TP1 CNN Challenge, deliver the best model!

first TP

By **Mariem ZAOUALI**
Posted on September 24, 2026

TP1 : CNN Challenge, deliver the best model!

Your goal is to optimize via transfer learning a deep learning model to achieve the best possible training and validation accuracy on the CIFAR-10 dataset, while minimizing both training and validation loss.

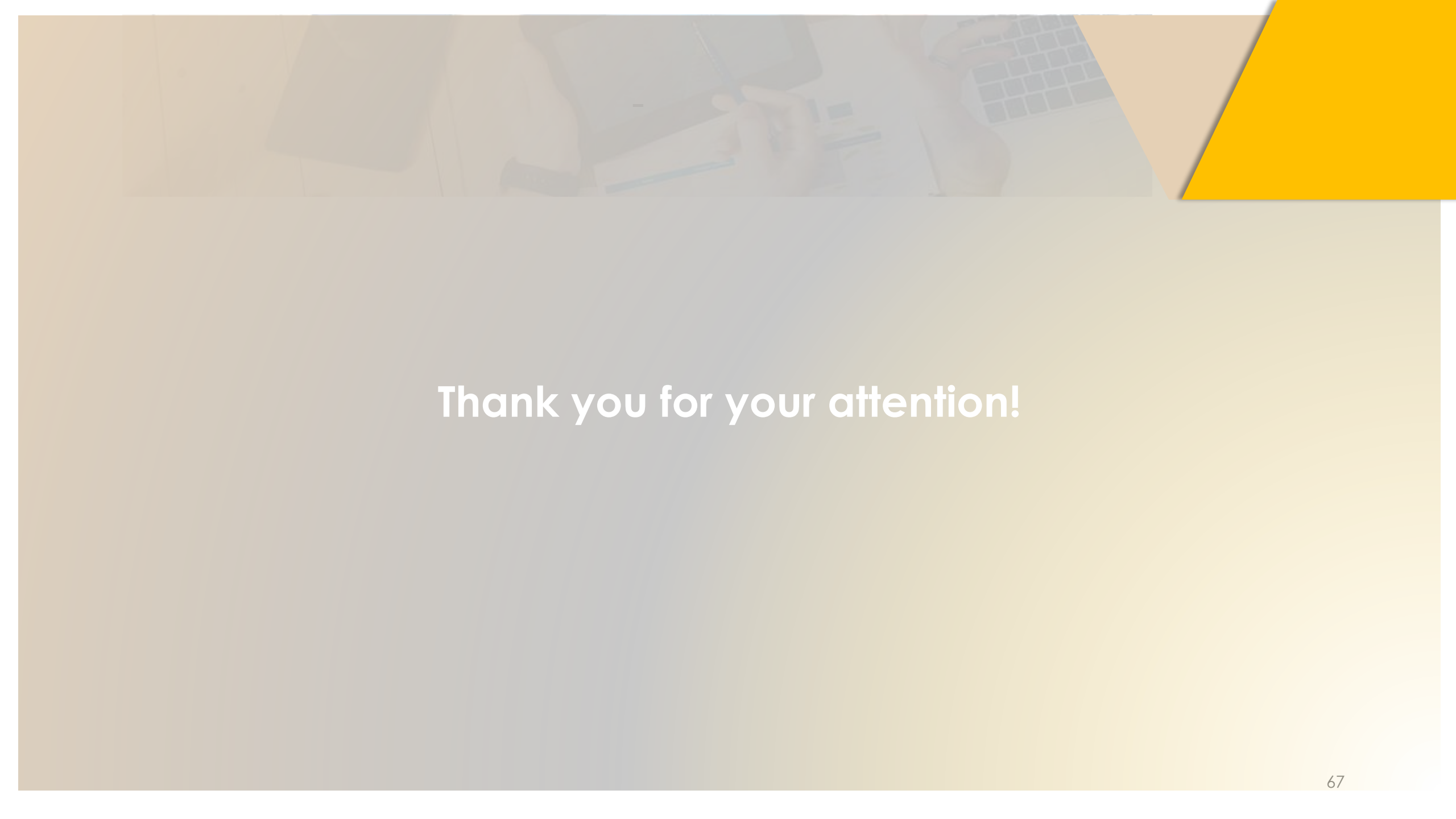
Dataset

- Dataset: CIFAR-10 URL: <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz>
Description: The CIFAR-10 dataset consists of 60,000 color images of size 32×32 pixels, divided into 10 classes (e.g., airplane, car, bird, cat, etc.), with 50,000 training images and 10,000 test images.

TP1 : CNN Challenge, deliver the best model!

- Dataset
- Task Description
- Model Design
- Training Strategy
- Evaluation
- Analysis
- Active Windows

Accédez aux paramètres pour activer Windows.



Thank you for your attention!